

Taller3B

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Escuela Politécnica Nacional

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Fecha: 20/01/2025

Repositorio: https://github.com/LuisALema/Metodos_Numericos_2025B/tree/main/Talleres/Ta

0.0.1 [Taller 3b] Métodos iterativos

Grafique la trayectoria de los siguientes sistemas de ecuaciones:

$$\begin{aligned}x_1 + x_2 &= 7 \\ -2x_1 + 5x_2 &= 0 \\ \bullet X_0 &= (0, 0) \\ \bullet X_0 &= (5, 2)\end{aligned}$$

$$\begin{aligned}x_1 + x_2 &= 6 \\ -2x_1 + x_2 &= 0\end{aligned}$$

Prueba 3 posiciones iniciales, encuentre una en la que diverge el sistema.

```
[2]: %load_ext autoreload
```

The autoreload extension is already loaded. To reload it, use:
%reload_ext autoreload

```
[3]: # %% Configuración inicial
%autoreload 2
from src import gauss_jacobi, gauss_seidel
import numpy as np
import matplotlib.pyplot as plt

# Función mejorada para graficar sistemas y trayectorias
```

```

def plot_system_with_trajectory(A, b, tray, title):
    tray_array = np.vstack([np.array(x).flatten() for x in tray])
    x_vals = np.linspace(min(tray_array[:, 0])-2, max(tray_array[:, 0])+2, 400)

    plt.figure(figsize=(12, 8))

    # Graficar cada ecuación del sistema
    colors = ['purple', 'orange']
    for i in range(A.shape[0]):
        if A[i, 1] != 0: # Para evitar división por cero
            y_vals = (b[i] - A[i, 0]*x_vals)/A[i, 1]
            plt.plot(x_vals, y_vals, color=colors[i], linestyle='--',
                     linewidth=2, alpha=0.7,
                     label=f'{A[i,0]}x + {A[i,1]}x = {b[i]}')

    # Graficar trayectoria
    plt.plot(tray_array[:, 0], tray_array[:, 1], 'b-', linewidth=1.5, alpha=0.
    ↪6, label='Trayectoria')
    plt.scatter(tray_array[:, 0], tray_array[:, 1], color='blue', s=30, alpha=0.
    ↪5)

    # Destacar puntos inicial y final
    plt.scatter(tray_array[0, 0], tray_array[0, 1], color='red', s=150,
                edgecolor='black', zorder=5, label=f'Inicio ({tray_array[0,0]:.
    ↪2f}, {tray_array[0,1]:.2f})')
    plt.scatter(tray_array[-1, 0], tray_array[-1, 1], color='green', s=150,
                edgecolor='black', zorder=5, label=f'Solución_
    ↪(({tray_array[-1,0]:.2f}, {tray_array[-1,1]:.2f})')

    plt.xlabel('x ', fontsize=12)
    plt.ylabel('x ', fontsize=12)
    plt.title(title, fontsize=14)
    plt.legend(loc='best', fontsize=10)
    plt.grid(True, linestyle='--', alpha=0.5)
    plt.axis('equal')
    plt.tight_layout()
    plt.show()

```

[01-21 11:20:14] [INFO] 2026-01-21 11:20:14.057909

[01-21 11:20:14] [INFO] 2026-01-21 11:20:14.396592

```

[4]: # %% Sistema 1 - GAUSS-JACOBI
A1 = np.array([[1, 1], [-2, 5]])
b1 = np.array([7, 0])
initial_positions = [np.array([5, 2]), np.array([0, 0]), np.array([-2, 4])]

print("=== SISTEMA 1 - MÉTODO DE GAUSS-JACOBI ===")

```

```

for i, x0 in enumerate(initial_positions, 1):
    print(f"\nCaso {i}: Posición inicial {x0}")
    try:
        x_j1, tray_j1 = gauss_jacobi(A=A1, b=b1, x0=x0, tol=1e-6, max_iter=50)
        plot_system_with_trajectory(A1, b1, tray_j1, f"Sistema 1 - Gauss-Jacobi (Inicial {i})")
        print(f"Solución encontrada: [{x_j1[0,0]:.6f}, {x_j1[1,0]:.6f}]")
    except Exception as e:
        print(f";Error! El método divergió: {str(e)}")

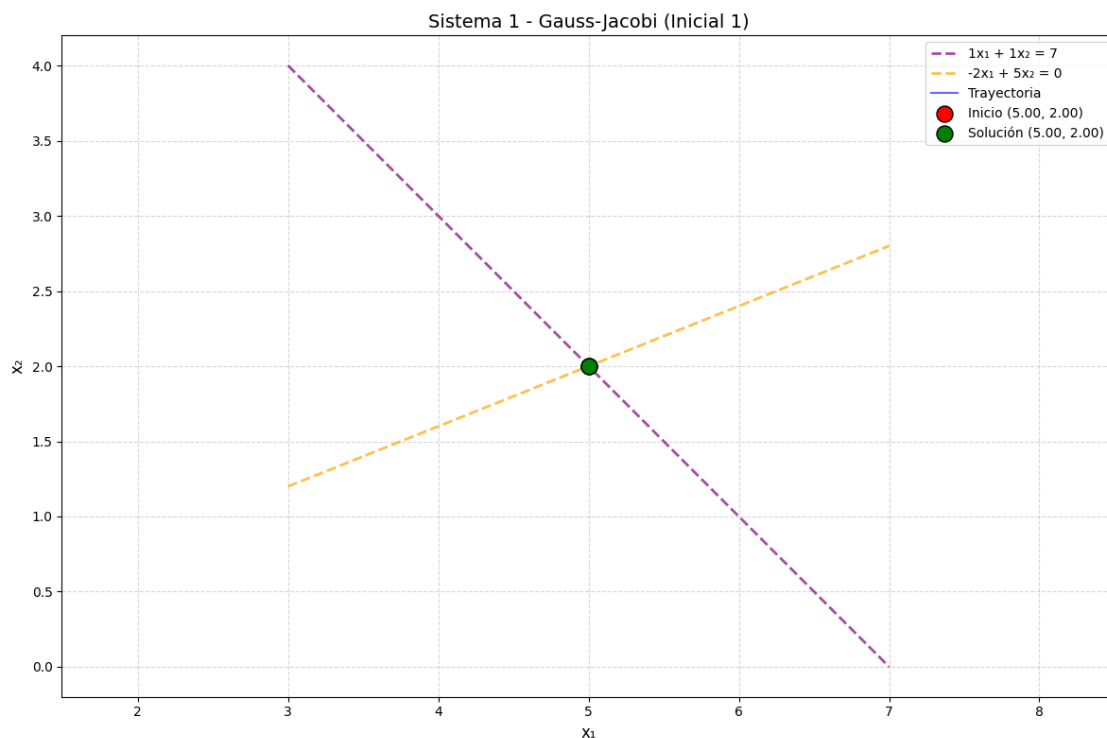
```

=== SISTEMA 1 - MÉTODO DE GAUSS-JACOBI ===

Caso 1: Posición inicial [5 2]

[01-21 11:20:23][INFO] i= 0 x: [5 2]

[01-21 11:20:23][INFO] i= 1 x: [[5. 2.]]



Solución encontrada: [5.000000, 2.000000]

Caso 2: Posición inicial [0 0]

[01-21 11:20:23][INFO] i= 0 x: [0 0]

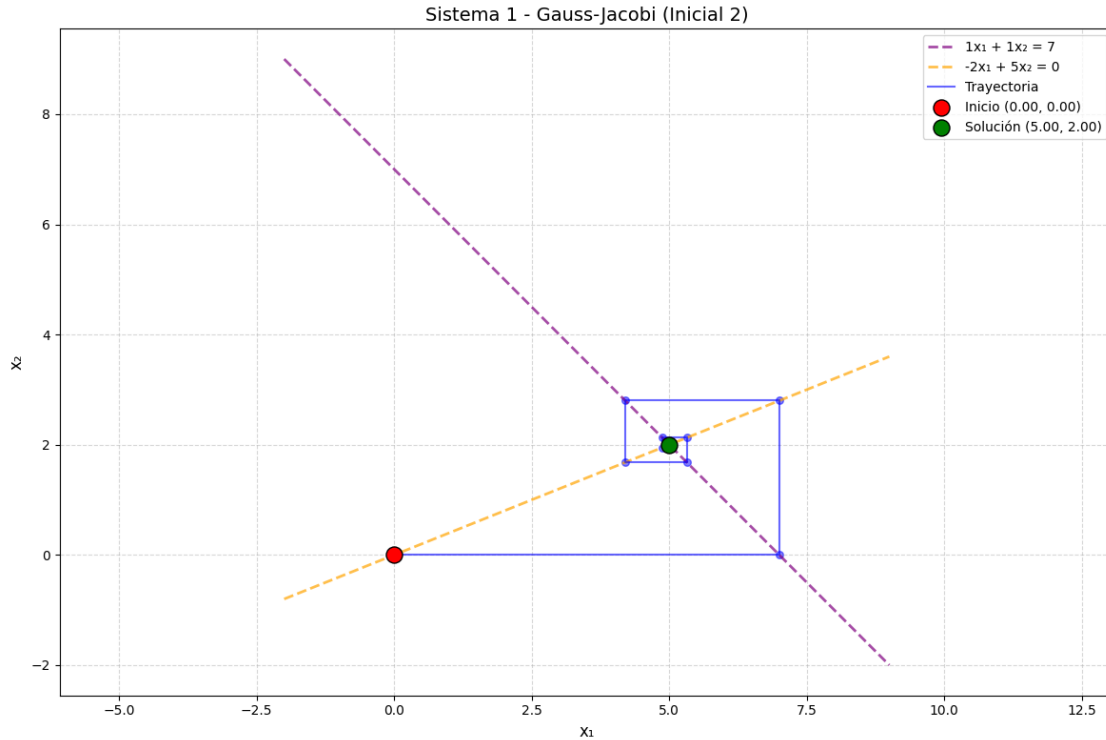
[01-21 11:20:23][INFO] i= 1 x: [[7. 0.]]

[01-21 11:20:23][INFO] i= 2 x: [[7. 2.8]]

[01-21 11:20:23][INFO] i= 3 x: [[4.2 2.8]]

[01-21 11:20:23][INFO] i= 4 x: [[4.2 1.68]]

[01-21 11:20:23][INFO] i= 5 x: [[5.32 1.68]]
[01-21 11:20:23][INFO] i= 6 x: [[5.32 2.128]]
[01-21 11:20:23][INFO] i= 7 x: [[4.872 2.128]]
[01-21 11:20:24][INFO] i= 8 x: [[4.872 1.9488]]
[01-21 11:20:24][INFO] i= 9 x: [[5.0512 1.9488]]
[01-21 11:20:24][INFO] i= 10 x: [[5.0512 2.02048]]
[01-21 11:20:24][INFO] i= 11 x: [[4.97952 2.02048]]
[01-21 11:20:24][INFO] i= 12 x: [[4.97952 1.991808]]
[01-21 11:20:24][INFO] i= 13 x: [[5.008192 1.991808]]
[01-21 11:20:24][INFO] i= 14 x: [[5.008192 2.0032768]]
[01-21 11:20:24][INFO] i= 15 x: [[4.9967232 2.0032768]]
[01-21 11:20:24][INFO] i= 16 x: [[4.9967232 1.99868928]]
[01-21 11:20:24][INFO] i= 17 x: [[5.00131072 1.99868928]]
[01-21 11:20:24][INFO] i= 18 x: [[5.00131072 2.00052429]]
[01-21 11:20:24][INFO] i= 19 x: [[4.99947571 2.00052429]]
[01-21 11:20:24][INFO] i= 20 x: [[4.99947571 1.99979028]]
[01-21 11:20:24][INFO] i= 21 x: [[5.00020972 1.99979028]]
[01-21 11:20:24][INFO] i= 22 x: [[5.00020972 2.00008389]]
[01-21 11:20:24][INFO] i= 23 x: [[4.99991611 2.00008389]]
[01-21 11:20:24][INFO] i= 24 x: [[4.99991611 1.99996645]]
[01-21 11:20:24][INFO] i= 25 x: [[5.00003355 1.99996645]]
[01-21 11:20:24][INFO] i= 26 x: [[5.00003355 2.00001342]]
[01-21 11:20:24][INFO] i= 27 x: [[4.99998658 2.00001342]]
[01-21 11:20:24][INFO] i= 28 x: [[4.99998658 1.99999463]]
[01-21 11:20:24][INFO] i= 29 x: [[5.00000537 1.99999463]]
[01-21 11:20:24][INFO] i= 30 x: [[5.00000537 2.00000215]]
[01-21 11:20:24][INFO] i= 31 x: [[4.99999785 2.00000215]]
[01-21 11:20:24][INFO] i= 32 x: [[4.99999785 1.99999914]]
[01-21 11:20:24][INFO] i= 33 x: [[5.00000086 1.99999914]]
[01-21 11:20:24][INFO] i= 34 x: [[5.00000086 2.00000034]]
[01-21 11:20:24][INFO] i= 35 x: [[4.99999966 2.00000034]]



Solución encontrada: [5.000000, 2.000000]

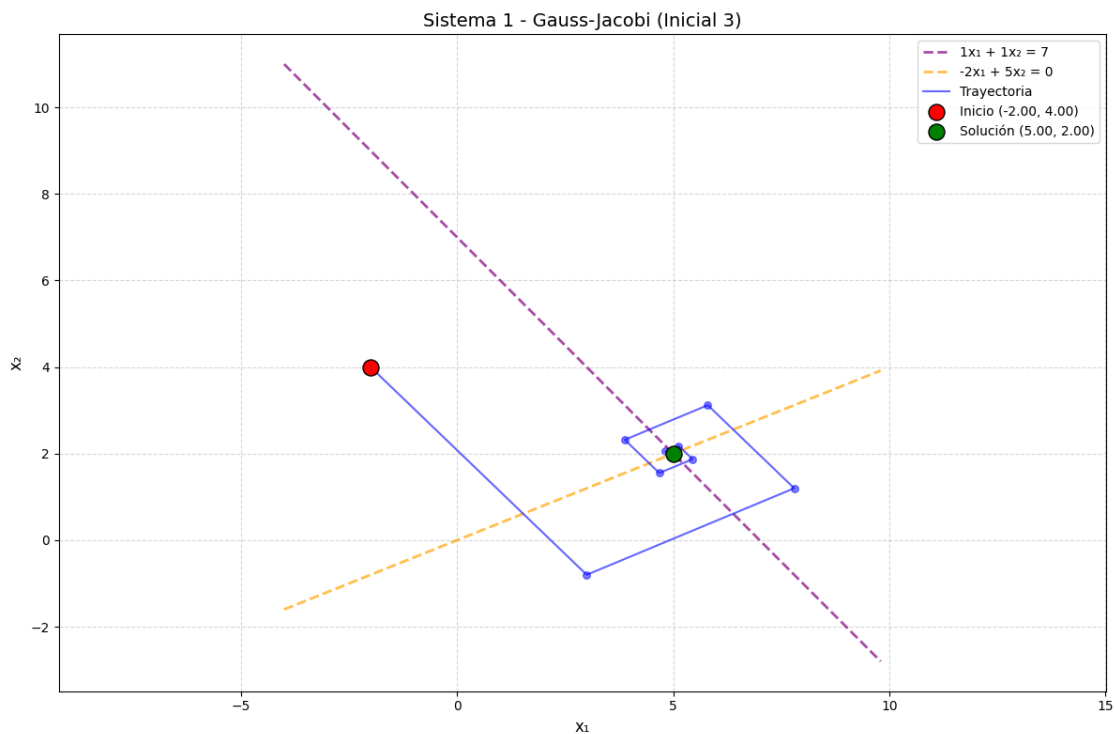
Caso 3: Posición inicial [-2 4]

```
[01-21 11:20:24] [INFO] i= 0 x: [-2  4]
[01-21 11:20:24] [INFO] i= 1 x: [[ 3.  -0.8]]
[01-21 11:20:24] [INFO] i= 2 x: [[7.8  1.2]]
[01-21 11:20:24] [INFO] i= 3 x: [[5.8  3.12]]
[01-21 11:20:24] [INFO] i= 4 x: [[3.88  2.32]]
[01-21 11:20:24] [INFO] i= 5 x: [[4.68  1.552]]
[01-21 11:20:24] [INFO] i= 6 x: [[5.448  1.872]]
[01-21 11:20:24] [INFO] i= 7 x: [[5.128  2.1792]]
[01-21 11:20:24] [INFO] i= 8 x: [[4.8208  2.0512]]
[01-21 11:20:24] [INFO] i= 9 x: [[4.9488  1.92832]]
[01-21 11:20:24] [INFO] i= 10 x: [[5.07168  1.97952]]
[01-21 11:20:24] [INFO] i= 11 x: [[5.02048  2.028672]]
[01-21 11:20:24] [INFO] i= 12 x: [[4.971328  2.008192]]
[01-21 11:20:24] [INFO] i= 13 x: [[4.991808  1.9885312]]
[01-21 11:20:24] [INFO] i= 14 x: [[5.0114688  1.9967232]]
[01-21 11:20:24] [INFO] i= 15 x: [[5.0032768  2.00458752]]
[01-21 11:20:24] [INFO] i= 16 x: [[4.99541248  2.00131072]]
[01-21 11:20:24] [INFO] i= 17 x: [[4.99868928  1.99816499]]
[01-21 11:20:24] [INFO] i= 18 x: [[5.00183501  1.99947571]]
[01-21 11:20:24] [INFO] i= 19 x: [[5.00052429  2.000734  ]]
[01-21 11:20:24] [INFO] i= 20 x: [[4.999266  2.00020972]]
```

```

[01-21 11:20:24] [INFO] i= 21 x: [[4.99979028 1.9997064 ]]
[01-21 11:20:24] [INFO] i= 22 x: [[5.0002936 1.99991611]]
[01-21 11:20:24] [INFO] i= 23 x: [[5.00008389 2.00011744]]
[01-21 11:20:24] [INFO] i= 24 x: [[4.99988256 2.00003355]]
[01-21 11:20:24] [INFO] i= 25 x: [[4.99996645 1.99995302]]
[01-21 11:20:24] [INFO] i= 26 x: [[5.00004698 1.99998658]]
[01-21 11:20:24] [INFO] i= 27 x: [[5.00001342 2.00001879]]
[01-21 11:20:24] [INFO] i= 28 x: [[4.99998121 2.00000537]]
[01-21 11:20:24] [INFO] i= 29 x: [[4.99999463 1.99999248]]
[01-21 11:20:24] [INFO] i= 30 x: [[5.00000752 1.99999785]]
[01-21 11:20:24] [INFO] i= 31 x: [[5.00000215 2.00000301]]
[01-21 11:20:24] [INFO] i= 32 x: [[4.99999699 2.00000086]]
[01-21 11:20:24] [INFO] i= 33 x: [[4.99999914 1.9999988 ]]
[01-21 11:20:24] [INFO] i= 34 x: [[5.0000012 1.99999966]]
[01-21 11:20:24] [INFO] i= 35 x: [[5.00000034 2.00000048]]

```



Solución encontrada: [5.000000, 2.000000]

```

[5]: # %% Sistema 1 - GAUSS-SEIDEL
print("\n=== SISTEMA 1 - MÉTODO DE GAUSS-SEIDEL ===")
for i, x0 in enumerate(initial_positions, 1):
    print(f"\nCaso {i}: Posición inicial {x0}")
    try:
        x_gs1, tray_gs1 = gauss_seidel(A=A1, b=b1, x0=x0, tol=1e-6, max_iter=50)

```

```

    plot_system_with_trajectory(A1, b1, tray_gs1, f"Sistema 1 - Gauss-Seidel (Inicial {i})")
    print(f"Solución encontrada: [{x_gs1[0,0]:.6f}, {x_gs1[1,0]:.6f}]")
except Exception as e:
    print(f"Error! El método divergió: {str(e)}")

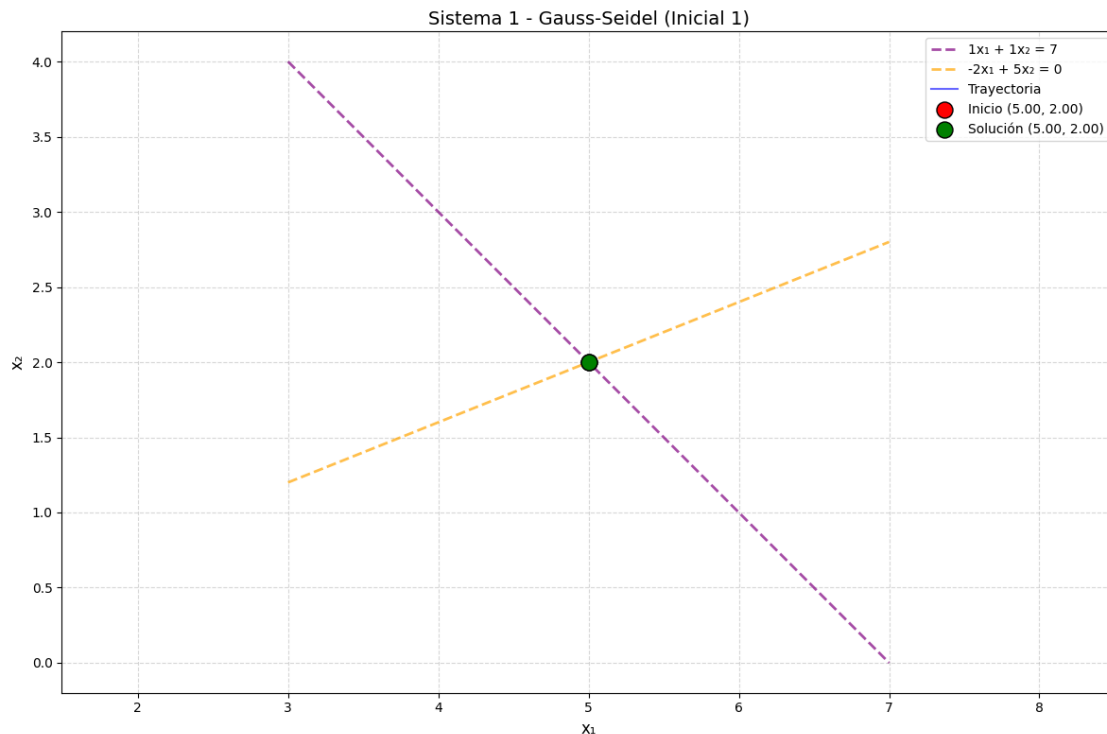
```

=== SISTEMA 1 - MÉTODO DE GAUSS-SEIDEL ===

Caso 1: Posición inicial [5 2]

[01-21 11:20:32][INFO] i= 0 x: [5 2]

[01-21 11:20:32][INFO] i= 1 x: [[5. 2.]]



Solución encontrada: [5.000000, 2.000000]

Caso 2: Posición inicial [0 0]

[01-21 11:20:32][INFO] i= 0 x: [0 0]

[01-21 11:20:32][INFO] i= 1 x: [[7. 2.8]]

[01-21 11:20:32][INFO] i= 2 x: [[4.2 1.68]]

[01-21 11:20:32][INFO] i= 3 x: [[5.32 2.128]]

[01-21 11:20:32][INFO] i= 4 x: [[4.872 1.9488]]

[01-21 11:20:32][INFO] i= 5 x: [[5.0512 2.02048]]

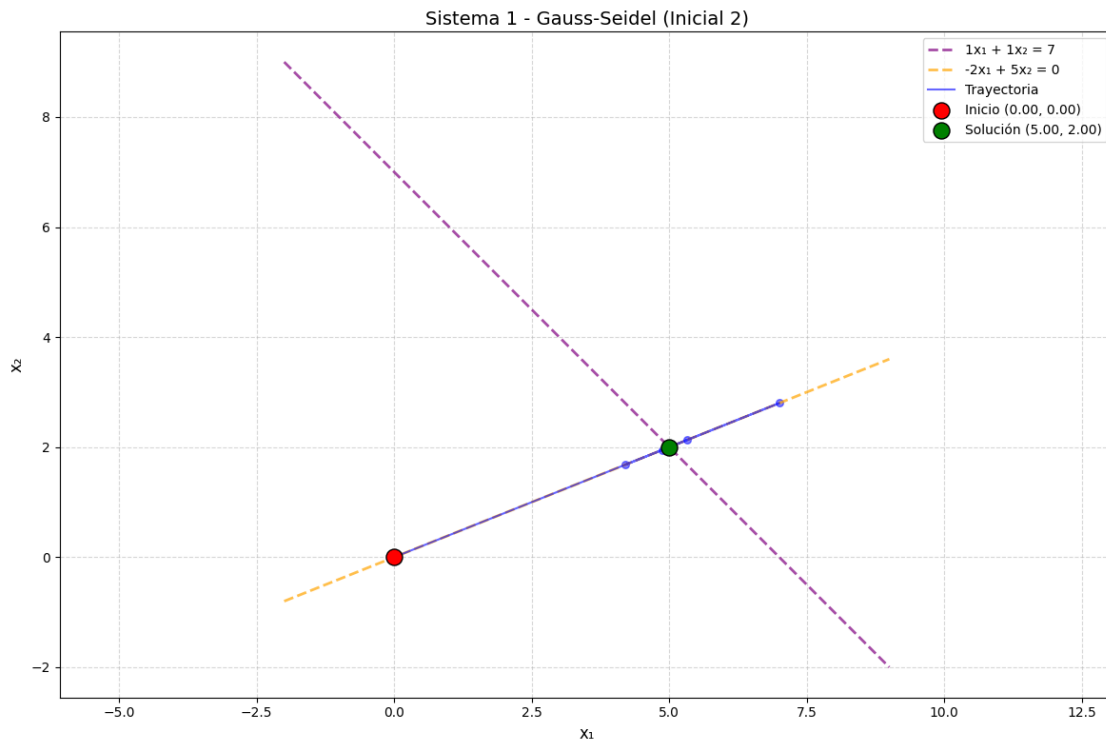
[01-21 11:20:32][INFO] i= 6 x: [[4.97952 1.991808]]

[01-21 11:20:32][INFO] i= 7 x: [[5.008192 2.0032768]]

```

[01-21 11:20:32][INFO] i= 8 x: [[4.9967232  1.99868928]]
[01-21 11:20:32][INFO] i= 9 x: [[5.00131072  2.00052429]]
[01-21 11:20:32][INFO] i= 10 x: [[4.99947571  1.99979028]]
[01-21 11:20:32][INFO] i= 11 x: [[5.00020972  2.00008389]]
[01-21 11:20:32][INFO] i= 12 x: [[4.99991611  1.99996645]]
[01-21 11:20:32][INFO] i= 13 x: [[5.00003355  2.00001342]]
[01-21 11:20:32][INFO] i= 14 x: [[4.99998658  1.99999463]]
[01-21 11:20:32][INFO] i= 15 x: [[5.00000537  2.00000215]]
[01-21 11:20:32][INFO] i= 16 x: [[4.99999785  1.99999914]]
[01-21 11:20:32][INFO] i= 17 x: [[5.00000086  2.00000034]]
[01-21 11:20:32][INFO] i= 18 x: [[4.99999966  1.99999986]]

```



Solución encontrada: [5.000000, 2.000000]

Caso 3: Posición inicial [-2 4]

```

[01-21 11:20:32][INFO] i= 0 x: [-2  4]
[01-21 11:20:32][INFO] i= 1 x: [[3.  1.2]]
[01-21 11:20:32][INFO] i= 2 x: [[5.8  2.32]]
[01-21 11:20:32][INFO] i= 3 x: [[4.68  1.872]]
[01-21 11:20:32][INFO] i= 4 x: [[5.128  2.0512]]
[01-21 11:20:32][INFO] i= 5 x: [[4.9488  1.97952]]
[01-21 11:20:32][INFO] i= 6 x: [[5.02048  2.008192]]
[01-21 11:20:32][INFO] i= 7 x: [[4.991808  1.9967232]]
[01-21 11:20:32][INFO] i= 8 x: [[5.0032768  2.00131072]]

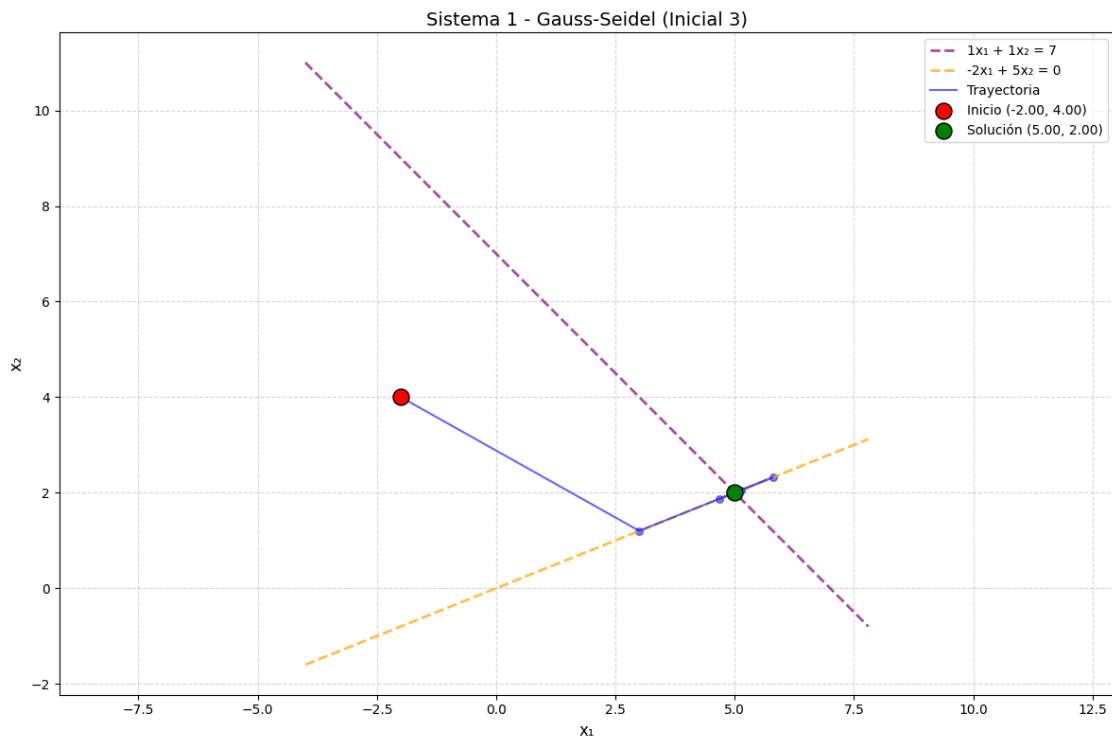
```



```

[01-21 11:20:32][INFO] i= 9 x: [[4.99868928 1.99947571]]
[01-21 11:20:32][INFO] i= 10 x: [[5.00052429 2.00020972]]
[01-21 11:20:32][INFO] i= 11 x: [[4.99979028 1.99991611]]
[01-21 11:20:32][INFO] i= 12 x: [[5.00008389 2.00003355]]
[01-21 11:20:32][INFO] i= 13 x: [[4.99996645 1.99998658]]
[01-21 11:20:32][INFO] i= 14 x: [[5.00001342 2.00000537]]
[01-21 11:20:32][INFO] i= 15 x: [[4.99999463 1.99999785]]
[01-21 11:20:32][INFO] i= 16 x: [[5.00000215 2.00000086]]
[01-21 11:20:32][INFO] i= 17 x: [[4.99999914 1.99999966]]
[01-21 11:20:32][INFO] i= 18 x: [[5.00000034 2.00000014]]

```



Solución encontrada: [5.000000, 2.000000]

```

[6]: # %% Sistema 2 - GAUSS-JACOBI
A2 = np.array([[1, 1], [-2, 1]])
b2 = np.array([6, 0])
initial_positions = [np.array([5, 2]), np.array([0, 0]), np.array([-2, 4])]

print("\n=== SISTEMA 2 - MÉTODO DE GAUSS-JACOBI ===")
for i, x0 in enumerate(initial_positions, 1):
    print(f"\nCaso {i}: Posición inicial {x0}")
    try:
        x_j2, tray_j2 = gauss_jacobi(A=A2, b=b2, x0=x0, tol=1e-6, max_iter=50)

```

```

        plot_system_with_trajectory(A2, b2, tray_j2, f"Sistema 2 - Gauss-Jacobi_
↪(Inicial {i})")
        print(f"Solución encontrada: [{x_j2[0,0]:.6f}, {x_j2[1,0]:.6f}]")
    except Exception as e:
        print(f";Error! El método divergió: {str(e)}")

```

=== SISTEMA 2 - MÉTODO DE GAUSS-JACOBI ===

Caso 1: Posición inicial [5 2]

```

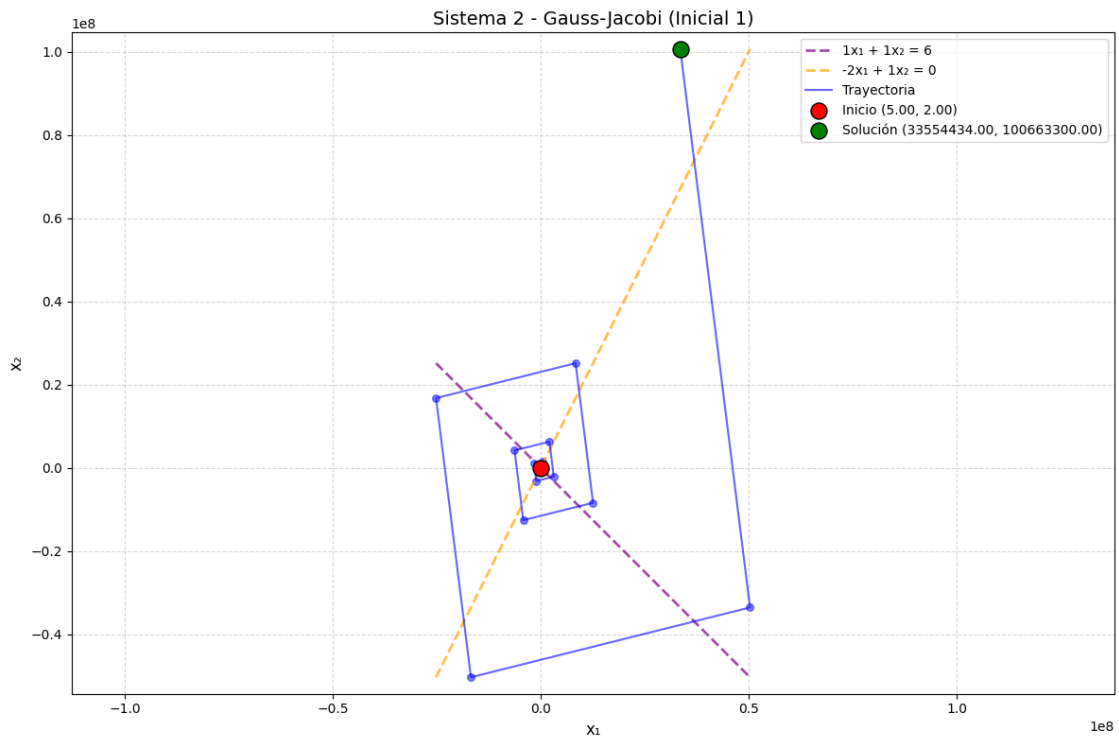
[01-21 11:20:37][INFO] i= 0 x: [5 2]
[01-21 11:20:37][INFO] i= 1 x: [[ 4. 10.]]
[01-21 11:20:37][INFO] i= 2 x: [[-4.  8.]]
[01-21 11:20:37][INFO] i= 3 x: [[-2. -8.]]
[01-21 11:20:37][INFO] i= 4 x: [[14. -4.]]
[01-21 11:20:37][INFO] i= 5 x: [[10. 28.]]
[01-21 11:20:37][INFO] i= 6 x: [[-22. 20.]]
[01-21 11:20:37][INFO] i= 7 x: [[-14. -44.]]
[01-21 11:20:37][INFO] i= 8 x: [[ 50. -28.]]
[01-21 11:20:37][INFO] i= 9 x: [[ 34. 100.]]
[01-21 11:20:37][INFO] i= 10 x: [[-94.  68.]]
[01-21 11:20:37][INFO] i= 11 x: [[ -62. -188.]]
[01-21 11:20:37][INFO] i= 12 x: [[ 194. -124.]]
[01-21 11:20:37][INFO] i= 13 x: [[130. 388.]]
[01-21 11:20:37][INFO] i= 14 x: [[-382. 260.]]
[01-21 11:20:37][INFO] i= 15 x: [[-254. -764.]]
[01-21 11:20:37][INFO] i= 16 x: [[ 770. -508.]]
[01-21 11:20:37][INFO] i= 17 x: [[ 514. 1540.]]
[01-21 11:20:37][INFO] i= 18 x: [[-1534. 1028.]]
[01-21 11:20:37][INFO] i= 19 x: [[-1022. -3068.]]
[01-21 11:20:37][INFO] i= 20 x: [[ 3074. -2044.]]
[01-21 11:20:37][INFO] i= 21 x: [[2050. 6148.]]
[01-21 11:20:37][INFO] i= 22 x: [[-6142. 4100.]]
[01-21 11:20:37][INFO] i= 23 x: [[ -4094. -12284.]]
[01-21 11:20:37][INFO] i= 24 x: [[12290. -8188.]]
[01-21 11:20:37][INFO] i= 25 x: [[ 8194. 24580.]]
[01-21 11:20:37][INFO] i= 26 x: [[-24574. 16388.]]
[01-21 11:20:37][INFO] i= 27 x: [[-16382. -49148.]]
[01-21 11:20:37][INFO] i= 28 x: [[ 49154. -32764.]]
[01-21 11:20:37][INFO] i= 29 x: [[32770. 98308.]]
[01-21 11:20:37][INFO] i= 30 x: [[-98302. 65540.]]
[01-21 11:20:37][INFO] i= 31 x: [[ -65534. -196604.]]
[01-21 11:20:37][INFO] i= 32 x: [[ 196610. -131068.]]
[01-21 11:20:37][INFO] i= 33 x: [[131074. 393220.]]
[01-21 11:20:37][INFO] i= 34 x: [[-393214. 262148.]]
[01-21 11:20:37][INFO] i= 35 x: [[-262142. -786428.]]
[01-21 11:20:37][INFO] i= 36 x: [[ 786434. -524284.]]
[01-21 11:20:37][INFO] i= 37 x: [[ 524290. 1572868.]]

```

```

[01-21 11:20:37][INFO] i= 38 x: [[-1572862.  1048580.]]
[01-21 11:20:37][INFO] i= 39 x: [[-1048574. -3145724.]]
[01-21 11:20:37][INFO] i= 40 x: [[ 3145730. -2097148.]]
[01-21 11:20:37][INFO] i= 41 x: [[2097154. 6291460.]]
[01-21 11:20:37][INFO] i= 42 x: [[-6291454.  4194308.]]
[01-21 11:20:37][INFO] i= 43 x: [[ -4194302. -12582908.]]
[01-21 11:20:37][INFO] i= 44 x: [[12582914. -8388604.]]
[01-21 11:20:37][INFO] i= 45 x: [[ 8388610. 25165828.]]
[01-21 11:20:37][INFO] i= 46 x: [[-25165822.  16777220.]]
[01-21 11:20:37][INFO] i= 47 x: [[-16777214. -50331644.]]
[01-21 11:20:37][INFO] i= 48 x: [[ 50331650. -33554428.]]
[01-21 11:20:37][INFO] i= 49 x: [[3.3554434e+07  1.0066330e+08]]

```



Solución encontrada: [33554434.000000, 100663300.000000]

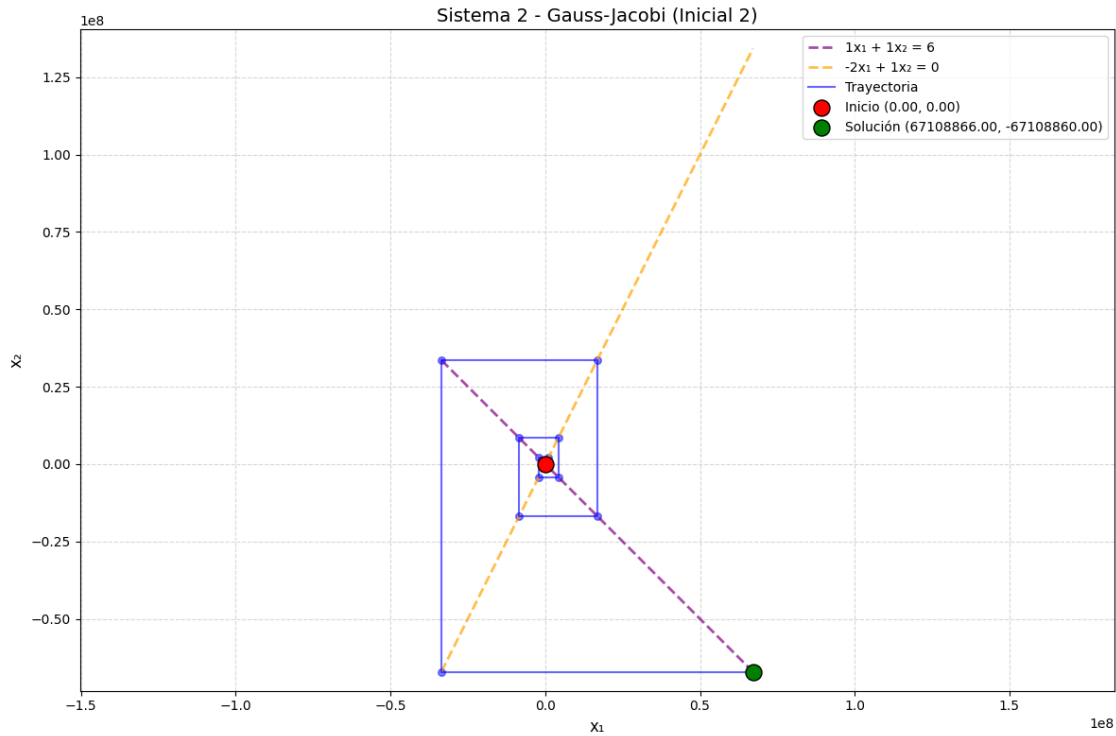
Caso 2: Posición inicial [0 0]

```

[01-21 11:20:37][INFO] i= 0 x: [0 0]
[01-21 11:20:37][INFO] i= 1 x: [[6. 0.]]
[01-21 11:20:37][INFO] i= 2 x: [[ 6. 12.]]
[01-21 11:20:37][INFO] i= 3 x: [[-6. 12.]]
[01-21 11:20:37][INFO] i= 4 x: [[ -6. -12.]]
[01-21 11:20:37][INFO] i= 5 x: [[ 18. -12.]]
[01-21 11:20:37][INFO] i= 6 x: [[18. 36.]]
[01-21 11:20:37][INFO] i= 7 x: [[-30. 36.]]

```

[01-21 11:20:37][INFO] i= 8 x: [[-30. -60.]]
[01-21 11:20:37][INFO] i= 9 x: [[66. -60.]]
[01-21 11:20:37][INFO] i= 10 x: [[66. 132.]]
[01-21 11:20:37][INFO] i= 11 x: [[-126. 132.]]
[01-21 11:20:37][INFO] i= 12 x: [[-126. -252.]]
[01-21 11:20:37][INFO] i= 13 x: [[258. -252.]]
[01-21 11:20:37][INFO] i= 14 x: [[258. 516.]]
[01-21 11:20:37][INFO] i= 15 x: [[-510. 516.]]
[01-21 11:20:37][INFO] i= 16 x: [[-510. -1020.]]
[01-21 11:20:37][INFO] i= 17 x: [[1026. -1020.]]
[01-21 11:20:37][INFO] i= 18 x: [[1026. 2052.]]
[01-21 11:20:37][INFO] i= 19 x: [[-2046. 2052.]]
[01-21 11:20:37][INFO] i= 20 x: [[-2046. -4092.]]
[01-21 11:20:37][INFO] i= 21 x: [[4098. -4092.]]
[01-21 11:20:37][INFO] i= 22 x: [[4098. 8196.]]
[01-21 11:20:37][INFO] i= 23 x: [[-8190. 8196.]]
[01-21 11:20:37][INFO] i= 24 x: [[-8190. -16380.]]
[01-21 11:20:37][INFO] i= 25 x: [[16386. -16380.]]
[01-21 11:20:37][INFO] i= 26 x: [[16386. 32772.]]
[01-21 11:20:37][INFO] i= 27 x: [[-32766. 32772.]]
[01-21 11:20:37][INFO] i= 28 x: [[-32766. -65532.]]
[01-21 11:20:37][INFO] i= 29 x: [[65538. -65532.]]
[01-21 11:20:37][INFO] i= 30 x: [[65538. 131076.]]
[01-21 11:20:37][INFO] i= 31 x: [[-131070. 131076.]]
[01-21 11:20:37][INFO] i= 32 x: [[-131070. -262140.]]
[01-21 11:20:37][INFO] i= 33 x: [[262146. -262140.]]
[01-21 11:20:37][INFO] i= 34 x: [[262146. 524292.]]
[01-21 11:20:37][INFO] i= 35 x: [[-524286. 524292.]]
[01-21 11:20:37][INFO] i= 36 x: [[-524286. -1048572.]]
[01-21 11:20:37][INFO] i= 37 x: [[1048578. -1048572.]]
[01-21 11:20:37][INFO] i= 38 x: [[1048578. 2097156.]]
[01-21 11:20:37][INFO] i= 39 x: [[-2097150. 2097156.]]
[01-21 11:20:37][INFO] i= 40 x: [[-2097150. -4194300.]]
[01-21 11:20:37][INFO] i= 41 x: [[4194306. -4194300.]]
[01-21 11:20:37][INFO] i= 42 x: [[4194306. 8388612.]]
[01-21 11:20:37][INFO] i= 43 x: [[-8388606. 8388612.]]
[01-21 11:20:37][INFO] i= 44 x: [[-8388606. -16777212.]]
[01-21 11:20:37][INFO] i= 45 x: [[16777218. -16777212.]]
[01-21 11:20:37][INFO] i= 46 x: [[16777218. 33554436.]]
[01-21 11:20:37][INFO] i= 47 x: [[-33554430. 33554436.]]
[01-21 11:20:37][INFO] i= 48 x: [[-33554430. -67108860.]]
[01-21 11:20:37][INFO] i= 49 x: [[67108866. -67108860.]]



Solución encontrada: [67108866.000000, -67108860.000000]

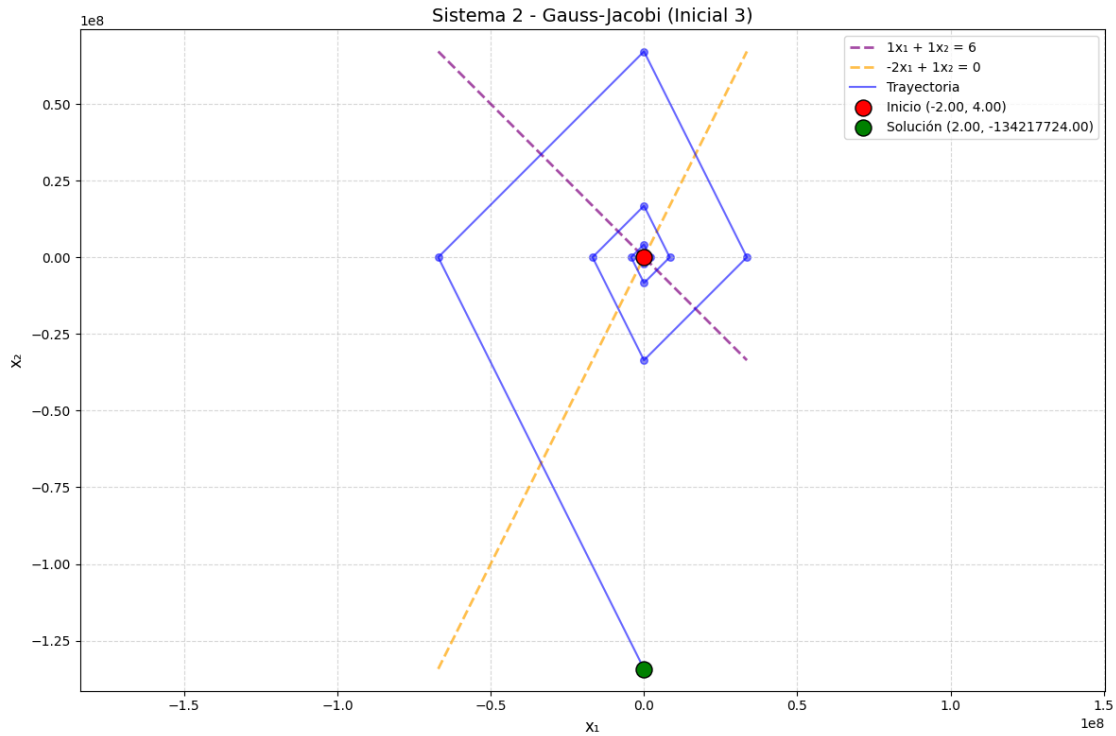
Caso 3: Posición inicial [-2 4]

```
[01-21 11:20:38][INFO] i= 0 x: [-2  4]
[01-21 11:20:38][INFO] i= 1 x: [[ 2. -4.]]
[01-21 11:20:38][INFO] i= 2 x: [[10.  4.]]
[01-21 11:20:38][INFO] i= 3 x: [[ 2. 20.]]
[01-21 11:20:38][INFO] i= 4 x: [[-14.  4.]]
[01-21 11:20:38][INFO] i= 5 x: [[ 2. -28.]]
[01-21 11:20:38][INFO] i= 6 x: [[34.  4.]]
[01-21 11:20:38][INFO] i= 7 x: [[ 2. 68.]]
[01-21 11:20:38][INFO] i= 8 x: [[-62.  4.]]
[01-21 11:20:38][INFO] i= 9 x: [[ 2. -124.]]
[01-21 11:20:38][INFO] i= 10 x: [[130.  4.]]
[01-21 11:20:38][INFO] i= 11 x: [[ 2. 260.]]
[01-21 11:20:38][INFO] i= 12 x: [[-254.  4.]]
[01-21 11:20:38][INFO] i= 13 x: [[ 2. -508.]]
[01-21 11:20:38][INFO] i= 14 x: [[514.  4.]]
[01-21 11:20:38][INFO] i= 15 x: [[ 2. 1028.]]
[01-21 11:20:38][INFO] i= 16 x: [[-1022.  4.]]
[01-21 11:20:38][INFO] i= 17 x: [[ 2.000e+00 -2.044e+03]]
[01-21 11:20:38][INFO] i= 18 x: [[2050.  4.]]
[01-21 11:20:38][INFO] i= 19 x: [[2.0e+00 4.1e+03]]
[01-21 11:20:38][INFO] i= 20 x: [[-4.094e+03 4.000e+00]]
```

```

[01-21 11:20:38][INFO] i= 21 x: [[ 2.000e+00 -8.188e+03]]
[01-21 11:20:38][INFO] i= 22 x: [[8.194e+03 4.000e+00]]
[01-21 11:20:38][INFO] i= 23 x: [[2.0000e+00 1.6388e+04]]
[01-21 11:20:38][INFO] i= 24 x: [[-1.6382e+04 4.0000e+00]]
[01-21 11:20:38][INFO] i= 25 x: [[ 2.0000e+00 -3.2764e+04]]
[01-21 11:20:38][INFO] i= 26 x: [[3.277e+04 4.000e+00]]
[01-21 11:20:38][INFO] i= 27 x: [[2.000e+00 6.554e+04]]
[01-21 11:20:38][INFO] i= 28 x: [[-6.5534e+04 4.0000e+00]]
[01-21 11:20:38][INFO] i= 29 x: [[ 2.00000e+00 -1.31068e+05]]
[01-21 11:20:38][INFO] i= 30 x: [[1.31074e+05 4.00000e+00]]
[01-21 11:20:38][INFO] i= 31 x: [[2.00000e+00 2.62148e+05]]
[01-21 11:20:38][INFO] i= 32 x: [[-2.62142e+05 4.00000e+00]]
[01-21 11:20:38][INFO] i= 33 x: [[ 2.00000e+00 -5.24284e+05]]
[01-21 11:20:38][INFO] i= 34 x: [[5.2429e+05 4.0000e+00]]
[01-21 11:20:38][INFO] i= 35 x: [[2.00000e+00 1.04858e+06]]
[01-21 11:20:38][INFO] i= 36 x: [[-1.048574e+06 4.000000e+00]]
[01-21 11:20:38][INFO] i= 37 x: [[ 2.000000e+00 -2.097148e+06]]
[01-21 11:20:38][INFO] i= 38 x: [[2.097154e+06 4.000000e+00]]
[01-21 11:20:38][INFO] i= 39 x: [[2.000000e+00 4.194308e+06]]
[01-21 11:20:38][INFO] i= 40 x: [[-4.194302e+06 4.000000e+00]]
[01-21 11:20:38][INFO] i= 41 x: [[ 2.000000e+00 -8.388604e+06]]
[01-21 11:20:38][INFO] i= 42 x: [[8.38861e+06 4.00000e+00]]
[01-21 11:20:38][INFO] i= 43 x: [[2.000000e+00 1.677722e+07]]
[01-21 11:20:38][INFO] i= 44 x: [[-1.6777214e+07 4.0000000e+00]]
[01-21 11:20:38][INFO] i= 45 x: [[ 2.0000000e+00 -3.3554428e+07]]
[01-21 11:20:38][INFO] i= 46 x: [[3.3554434e+07 4.0000000e+00]]
[01-21 11:20:38][INFO] i= 47 x: [[2.0000000e+00 6.7108868e+07]]
[01-21 11:20:38][INFO] i= 48 x: [[-6.7108862e+07 4.0000000e+00]]
[01-21 11:20:38][INFO] i= 49 x: [[ 2.00000000e+00 -1.34217724e+08]]

```



Solución encontrada: [2.000000, -134217724.000000]

```
[7]: # %% Sistema 2 - GAUSS-SEIDEL
print("\n=== SISTEMA 2 - MÉTODO DE GAUSS-SEIDEL ===")
for i, x0 in enumerate(initial_positions, 1):
    print(f"\nCaso {i}: Posición inicial {x0}")
    try:
        x_gs2, tray_gs2 = gauss_seidel(A=A2, b=b2, x0=x0, tol=1e-6, max_iter=50)
        plot_system_with_trajectory(A2, b2, tray_gs2, f"Sistema 2 - \u2192Gauss-Seidel (Inicial {i})")
        print(f"Solución encontrada: [{x_gs2[0,0]:.6f}, {x_gs2[1,0]:.6f}]")
    except Exception as e:
        print(f";Error! El método divergió: {str(e)}")
```

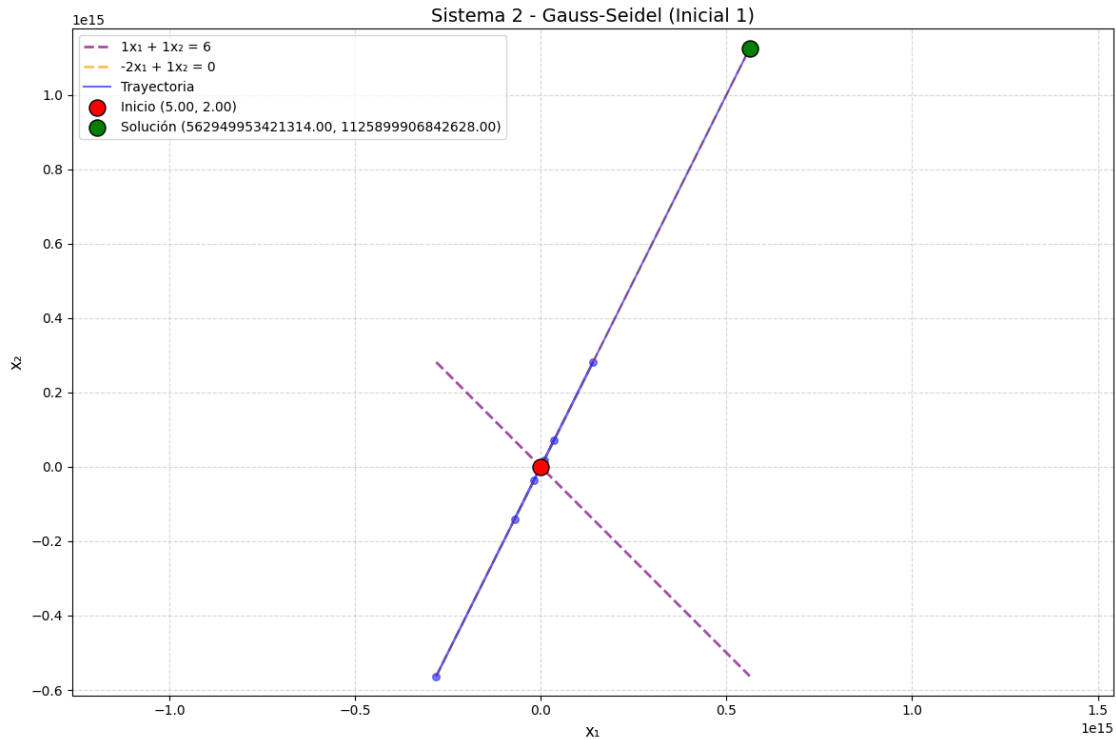
=== SISTEMA 2 - MÉTODO DE GAUSS-SEIDEL ===

```
Caso 1: Posición inicial [5 2]
[01-21 11:20:53][INFO] i= 0 x: [5 2]
[01-21 11:20:53][INFO] i= 1 x: [[4. 8.]]
[01-21 11:20:53][INFO] i= 2 x: [[-2. -4.]]
[01-21 11:20:53][INFO] i= 3 x: [[10. 20.]]
[01-21 11:20:53][INFO] i= 4 x: [[-14. -28.]]
[01-21 11:20:53][INFO] i= 5 x: [[34. 68.]]
```

```

[01-21 11:20:53][INFO] i= 6 x: [[ -62. -124.]]
[01-21 11:20:53][INFO] i= 7 x: [[130. 260.]]
[01-21 11:20:53][INFO] i= 8 x: [[-254. -508.]]
[01-21 11:20:53][INFO] i= 9 x: [[ 514. 1028.]]
[01-21 11:20:53][INFO] i= 10 x: [[-1022. -2044.]]
[01-21 11:20:53][INFO] i= 11 x: [[2050. 4100.]]
[01-21 11:20:53][INFO] i= 12 x: [[-4094. -8188.]]
[01-21 11:20:53][INFO] i= 13 x: [[ 8194. 16388.]]
[01-21 11:20:53][INFO] i= 14 x: [[-16382. -32764.]]
[01-21 11:20:53][INFO] i= 15 x: [[32770. 65540.]]
[01-21 11:20:53][INFO] i= 16 x: [[ -65534. -131068.]]
[01-21 11:20:53][INFO] i= 17 x: [[131074. 262148.]]
[01-21 11:20:53][INFO] i= 18 x: [[-262142. -524284.]]
[01-21 11:20:53][INFO] i= 19 x: [[ 524290. 1048580.]]
[01-21 11:20:53][INFO] i= 20 x: [[-1048574. -2097148.]]
[01-21 11:20:53][INFO] i= 21 x: [[2097154. 4194308.]]
[01-21 11:20:53][INFO] i= 22 x: [[-4194302. -8388604.]]
[01-21 11:20:53][INFO] i= 23 x: [[ 8388610. 16777220.]]
[01-21 11:20:53][INFO] i= 24 x: [[-16777214. -33554428.]]
[01-21 11:20:53][INFO] i= 25 x: [[33554434. 67108868.]]
[01-21 11:20:53][INFO] i= 26 x: [[-6.71088620e+07 -1.34217724e+08]]
[01-21 11:20:53][INFO] i= 27 x: [[1.3421773e+08 2.6843546e+08]]
[01-21 11:20:53][INFO] i= 28 x: [[-2.68435454e+08 -5.36870908e+08]]
[01-21 11:20:53][INFO] i= 29 x: [[5.36870914e+08 1.07374183e+09]]
[01-21 11:20:53][INFO] i= 30 x: [[-1.07374182e+09 -2.14748364e+09]]
[01-21 11:20:53][INFO] i= 31 x: [[2.14748365e+09 4.29496730e+09]]
[01-21 11:20:53][INFO] i= 32 x: [[-4.29496729e+09 -8.58993459e+09]]
[01-21 11:20:53][INFO] i= 33 x: [[8.58993459e+09 1.71798692e+10]]
[01-21 11:20:53][INFO] i= 34 x: [[-1.71798692e+10 -3.43597384e+10]]
[01-21 11:20:53][INFO] i= 35 x: [[3.43597384e+10 6.87194767e+10]]
[01-21 11:20:53][INFO] i= 36 x: [[-6.87194767e+10 -1.37438953e+11]]
[01-21 11:20:53][INFO] i= 37 x: [[1.37438953e+11 2.74877907e+11]]
[01-21 11:20:53][INFO] i= 38 x: [[-2.74877907e+11 -5.49755814e+11]]
[01-21 11:20:53][INFO] i= 39 x: [[5.49755814e+11 1.09951163e+12]]
[01-21 11:20:53][INFO] i= 40 x: [[-1.09951163e+12 -2.19902326e+12]]
[01-21 11:20:53][INFO] i= 41 x: [[2.19902326e+12 4.39804651e+12]]
[01-21 11:20:53][INFO] i= 42 x: [[-4.39804651e+12 -8.79609302e+12]]
[01-21 11:20:53][INFO] i= 43 x: [[8.79609302e+12 1.75921860e+13]]
[01-21 11:20:53][INFO] i= 44 x: [[-1.75921860e+13 -3.51843721e+13]]
[01-21 11:20:53][INFO] i= 45 x: [[3.51843721e+13 7.03687442e+13]]
[01-21 11:20:53][INFO] i= 46 x: [[-7.03687442e+13 -1.40737488e+14]]
[01-21 11:20:53][INFO] i= 47 x: [[1.40737488e+14 2.81474977e+14]]
[01-21 11:20:53][INFO] i= 48 x: [[-2.81474977e+14 -5.62949953e+14]]
[01-21 11:20:53][INFO] i= 49 x: [[5.62949953e+14 1.12589991e+15]]

```

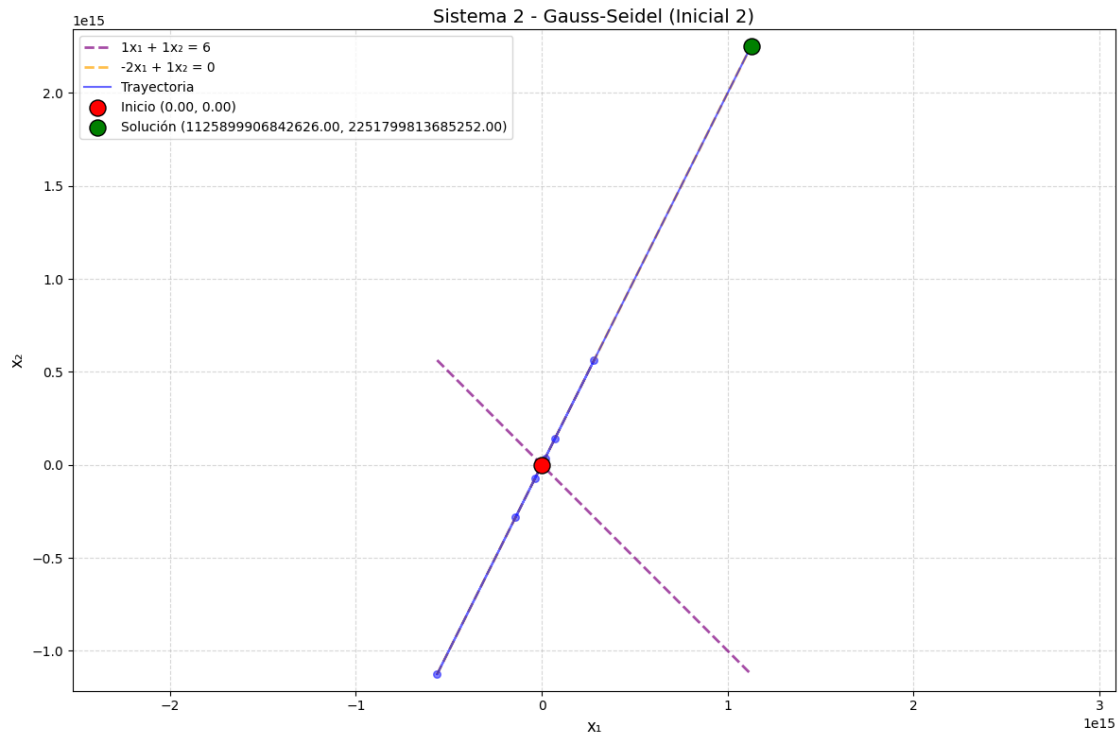



Solución encontrada: [562949953421314.000000, 1125899906842628.000000]

Caso 2: Posición inicial [0 0]

```
[01-21 11:20:53][INFO] i= 0 x: [0 0]
[01-21 11:20:53][INFO] i= 1 x: [[ 6. 12.]]
[01-21 11:20:53][INFO] i= 2 x: [[ -6. -12.]]
[01-21 11:20:53][INFO] i= 3 x: [[18. 36.]]
[01-21 11:20:53][INFO] i= 4 x: [[-30. -60.]]
[01-21 11:20:53][INFO] i= 5 x: [[ 66. 132.]]
[01-21 11:20:53][INFO] i= 6 x: [[-126. -252.]]
[01-21 11:20:53][INFO] i= 7 x: [[258. 516.]]
[01-21 11:20:53][INFO] i= 8 x: [[ -510. -1020.]]
[01-21 11:20:53][INFO] i= 9 x: [[1026. 2052.]]
[01-21 11:20:53][INFO] i= 10 x: [[-2046. -4092.]]
[01-21 11:20:53][INFO] i= 11 x: [[4098. 8196.]]
[01-21 11:20:53][INFO] i= 12 x: [[ -8190. -16380.]]
[01-21 11:20:53][INFO] i= 13 x: [[16386. 32772.]]
[01-21 11:20:53][INFO] i= 14 x: [[-32766. -65532.]]
[01-21 11:20:53][INFO] i= 15 x: [[ 65538. 131076.]]
[01-21 11:20:53][INFO] i= 16 x: [[-131070. -262140.]]
[01-21 11:20:53][INFO] i= 17 x: [[262146. 524292.]]
[01-21 11:20:53][INFO] i= 18 x: [[ -524286. -1048572.]]
[01-21 11:20:53][INFO] i= 19 x: [[1048578. 2097156.]]
[01-21 11:20:53][INFO] i= 20 x: [[-2097150. -4194300.]]
```

[01-21 11:20:53][INFO] i= 21 x: [[4194306. 8388612.]]
[01-21 11:20:53][INFO] i= 22 x: [[-8388606. -16777212.]]
[01-21 11:20:53][INFO] i= 23 x: [[16777218. 33554436.]]
[01-21 11:20:53][INFO] i= 24 x: [[-33554430. -67108860.]]
[01-21 11:20:53][INFO] i= 25 x: [[6.71088660e+07 1.34217732e+08]]
[01-21 11:20:53][INFO] i= 26 x: [[-1.34217726e+08 -2.68435452e+08]]
[01-21 11:20:53][INFO] i= 27 x: [[2.68435458e+08 5.36870916e+08]]
[01-21 11:20:53][INFO] i= 28 x: [[-5.36870910e+08 -1.07374182e+09]]
[01-21 11:20:53][INFO] i= 29 x: [[1.07374183e+09 2.14748365e+09]]
[01-21 11:20:53][INFO] i= 30 x: [[-2.14748365e+09 -4.29496729e+09]]
[01-21 11:20:53][INFO] i= 31 x: [[4.2949673e+09 8.5899346e+09]]
[01-21 11:20:53][INFO] i= 32 x: [[-8.58993459e+09 -1.71798692e+10]]
[01-21 11:20:53][INFO] i= 33 x: [[1.71798692e+10 3.43597384e+10]]
[01-21 11:20:53][INFO] i= 34 x: [[-3.43597384e+10 -6.87194767e+10]]
[01-21 11:20:53][INFO] i= 35 x: [[6.87194767e+10 1.37438953e+11]]
[01-21 11:20:53][INFO] i= 36 x: [[-1.37438953e+11 -2.74877907e+11]]
[01-21 11:20:53][INFO] i= 37 x: [[2.74877907e+11 5.49755814e+11]]
[01-21 11:20:53][INFO] i= 38 x: [[-5.49755814e+11 -1.09951163e+12]]
[01-21 11:20:53][INFO] i= 39 x: [[1.09951163e+12 2.19902326e+12]]
[01-21 11:20:53][INFO] i= 40 x: [[-2.19902326e+12 -4.39804651e+12]]
[01-21 11:20:53][INFO] i= 41 x: [[4.39804651e+12 8.79609302e+12]]
[01-21 11:20:53][INFO] i= 42 x: [[-8.79609302e+12 -1.75921860e+13]]
[01-21 11:20:53][INFO] i= 43 x: [[1.75921860e+13 3.51843721e+13]]
[01-21 11:20:53][INFO] i= 44 x: [[-3.51843721e+13 -7.03687442e+13]]
[01-21 11:20:53][INFO] i= 45 x: [[7.03687442e+13 1.40737488e+14]]
[01-21 11:20:53][INFO] i= 46 x: [[-1.40737488e+14 -2.81474977e+14]]
[01-21 11:20:53][INFO] i= 47 x: [[2.81474977e+14 5.62949953e+14]]
[01-21 11:20:53][INFO] i= 48 x: [[-5.62949953e+14 -1.12589991e+15]]
[01-21 11:20:53][INFO] i= 49 x: [[1.12589991e+15 2.25179981e+15]]

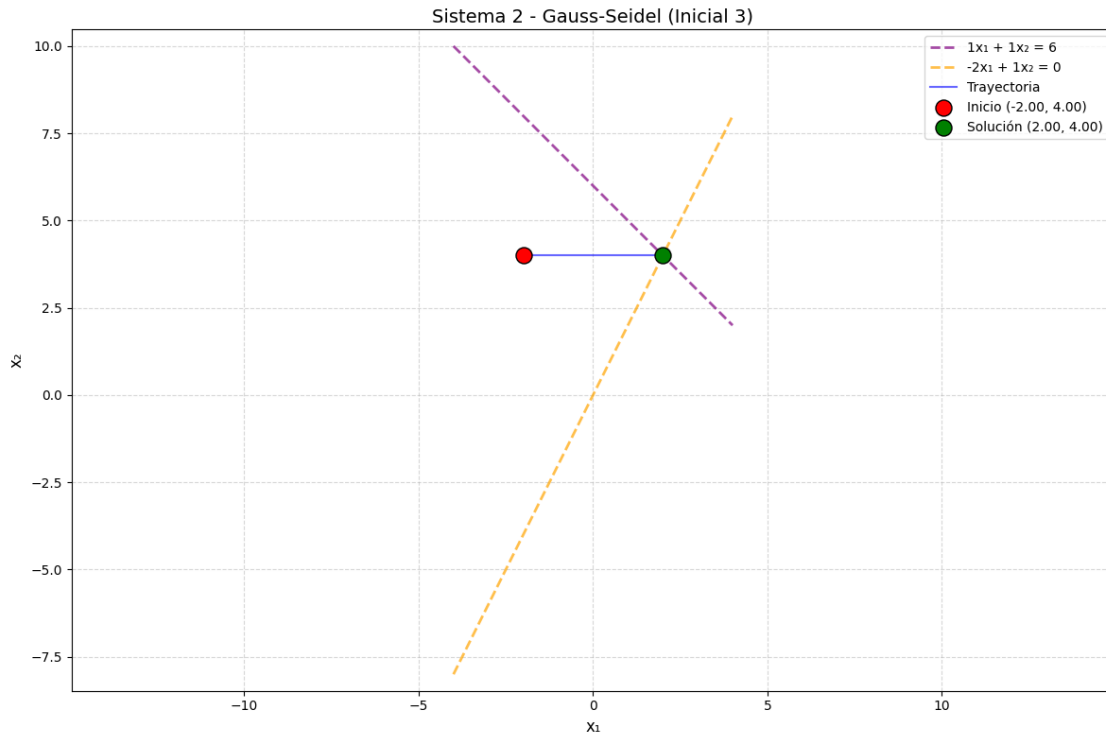


Solución encontrada: [1125899906842626.000000, 2251799813685252.000000]

Caso 3: Posición inicial [-2 4]

[01-21 11:20:53] [INFO] i= 0 x: [-2 4]

[01-21 11:20:53] [INFO] i= 1 x: [[2. 4.]]



Solución encontrada: [2.000000, 4.000000]

```
[9]: import numpy as np
import matplotlib.pyplot as plt

def plot_error_convergence(trayectoria, titulo="Convergencia (Norma del  $\ell_1$  error)"):
    """
    Grafica la norma del cambio entre iteraciones:
         $e_k = ||x^{(k)} - x^{(k-1)}||$ 
    usando escala logarítmica en Y.
    """
    X = []
    for xk in trayectoria:
        xk = np.array(xk, dtype=float).reshape(-1) # (n,1) o (n,) -> (n,)
        X.append(xk)
    X = np.array(X)

    if len(X) < 2:
        print("No hay suficientes iteraciones para graficar.")
        return

    errores = []
```

```

for k in range(1, len(X)):
    errores.append(np.linalg.norm(X[k] - X[k-1]))

plt.figure(figsize=(7, 4))
plt.plot(range(1, len(errores) + 1), errores, marker='o')
plt.yscale('log')
plt.xlabel("Iteración")
plt.ylabel(r"$\|x^{\{k\}} - x^{\{k-1\}}\|$")
plt.title(titulo)
plt.grid(True)
plt.show()

# %% Ejercicio 3 - GAUSS-JACOBI (4x4)
A3 = np.array([
    [ 1, -1,  2,  0],
    [-1,  1, -1,  3],
    [ 2, -1,  1, -1],
    [ 0,  3, -1,  2]
], dtype=float)

b3 = np.array([1360, 1130, 1350, 3650], dtype=float)

initial_positions = [
    np.array([0, 0, 0, 0], dtype=float),
    np.array([1, 1, 1, 1], dtype=float),
    np.array([1000, 1000, 1000, 1000], dtype=float)
]

print("\n=== EJERCICIO 3 - MÉTODO DE GAUSS-JACOBI ===")
for i, x0 in enumerate(initial_positions, 1):
    print(f"\nCaso {i}: Posición inicial {x0}")
    try:
        x_j3, tray_j3 = gauss_jacobi(A=A3, b=b3, x0=x0, tol=1e-6, max_iter=50)

        # "Trayectoria" para 4D: convergencia del error
        plot_error_convergence(
            tray_j3,
            titulo=f"Ejercicio 3 - Gauss-Jacobi (Caso {i})"
        )

        print(
            "Solución encontrada: "
            f"[{x_j3[0,0]:.6f}, {x_j3[1,0]:.6f}, "
            f"{x_j3[2,0]:.6f}, {x_j3[3,0]:.6f}]"
        )
    except Exception as e:

```

```
print(f";Error! El método divergió: {str(e)}")
```

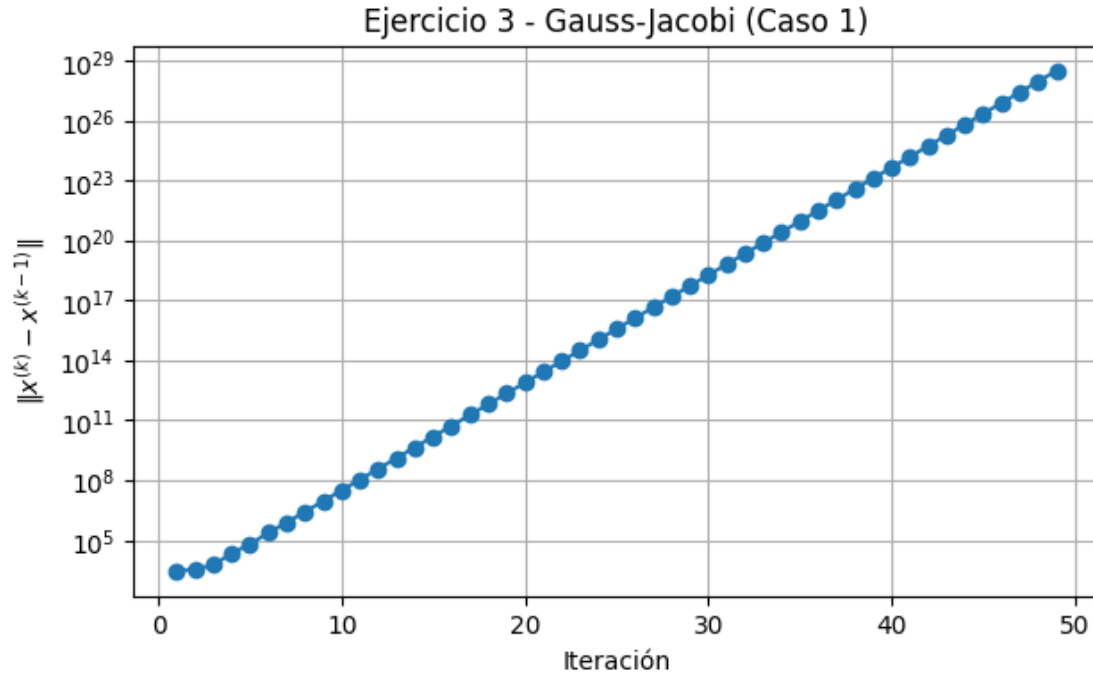
=== EJERCICIO 3 - MÉTODO DE GAUSS-JACOBI ===

Caso 1: Posición inicial [0. 0. 0. 0.]

```
[01-21 11:21:10][INFO] i= 0 x: [0. 0. 0. 0.]
[01-21 11:21:10][INFO] i= 1 x: [[1360. 1130. 1350. 1825.]]
[01-21 11:21:10][INFO] i= 2 x: [[ -210. -1635. 1585. 805.]]
[01-21 11:21:10][INFO] i= 3 x: [[-3445. 90. 940. 5070.]]
[01-21 11:21:10][INFO] i= 4 x: [[ -430. -16585. 13400. 2160.]]
[01-21 11:21:10][INFO] i= 5 x: [[-42025. 7620. -12215. 33402.5]]
[01-21 11:21:10][INFO] i= 6 x: [[ 33410. -153317.5 126422.5 -15712.5]]
[01-21 11:21:10][INFO] i= 7 x: [[-404802.5 208100. -234500. 295012.5]]
[01-21 11:21:10][INFO] i= 8 x: [[ 678460. -1523210. 1314067.5 -427575. ]]
[01-21 11:21:10][INFO] i= 9 x: [[-4149985. 3276382.5 -3306355.
2943673.75]]
[01-21 11:21:10][INFO] i= 10 x: [[ 9890452.5 -16286231.25 14521376.25
-6565926.25]]
[01-21 11:21:10][INFO] i= 11 x: [[-45327623.75 44110737.5 -42631712.5
31691860. ]]
[01-21 11:21:10][INFO] i= 12 x: [[ 1.29375522e+08 -1.83033786e+08
1.66459195e+08 -8.74801375e+07]]
[01-21 11:21:10][INFO] i= 13 x: [[-5.15950816e+08 5.58276260e+08
-5.29263619e+08 3.57782102e+08]]
[01-21 11:21:10][INFO] i= 14 x: [[ 1.61680486e+09 -2.11855961e+09
1.94796134e+09 -1.10204437e+09]]
[01-21 11:21:10][INFO] i= 15 x: [[-6.01448094e+09 6.87090046e+09
-6.45421235e+09 4.15182191e+09]]
[01-21 11:21:10][INFO] i= 16 x: [[ 1.97793265e+10 -2.49241579e+10
2.30516856e+10 -1.35334550e+10]]
[01-21 11:21:10][INFO] i= 17 x: [[-7.10275277e+10 8.34313783e+10
-7.80162646e+10 4.89120815e+10]]
[01-21 11:21:10][INFO] i= 18 x: [[ 2.39463909e+11 -2.95780036e+11
2.74398517e+11 -1.64155198e+11]]
[01-21 11:21:10][INFO] i= 19 x: [[-8.44577068e+11 1.00632802e+12
-9.38863050e+11 5.80869314e+11]]
[01-21 11:21:10][INFO] i= 20 x: [[ 2.88405412e+12 -3.52604806e+12
3.27635147e+12 -1.97892355e+12]]
[01-21 11:21:10][INFO] i= 21 x: [[-1.00787510e+13 1.20971763e+13
-1.12730799e+13 6.92724782e+12]]
[01-21 11:21:10][INFO] i= 22 x: [[ 3.46433360e+13 -4.21335743e+13
3.91819261e+13 -2.37823043e+13]]
[01-21 11:21:10][INFO] i= 23 x: [[-1.20497426e+14 1.45172175e+14
-1.35202551e+14 8.27913245e+13]]
[01-21 11:21:10][INFO] i= 24 x: [[ 4.15577276e+14 -5.04073951e+14
4.68958352e+14 -2.85359538e+14]]
[01-21 11:21:10][INFO] i= 25 x: [[-1.44199066e+15 1.74061424e+15
```

-1.62058804e+15 9.90590102e+14]]
 [01-21 11:21:10][INFO] i= 26 x: [[4.98179032e+15 -6.03434900e+15
 5.61518565e+15 -3.42121538e+15]]
 [01-21 11:21:10][INFO] i= 27 x: [[-1.72647203e+16 2.08606221e+16
 -1.94191450e+16 1.18591163e+16]]
 [01-21 11:21:10][INFO] i= 28 x: [[5.96989122e+16 -7.22612143e+16
 6.72491791e+16 -4.10005057e+16]]
 [01-21 11:21:10][INFO] i= 29 x: [[-2.06759572e+17 2.49949608e+17
 -2.32659544e+17 1.42016411e+17]]
 [01-21 11:21:10][INFO] i= 30 x: [[7.15268697e+17 -8.65468350e+17
 8.05485164e+17 -4.91254185e+17]]
 [01-21 11:21:10][INFO] i= 31 x: [[-2.47643868e+18 2.99451642e+18
 -2.78725993e+18 1.70094511e+18]]
 [01-21 11:21:10][INFO] i= 32 x: [[8.56903627e+18 -1.03665339e+19
 9.64833888e+18 -5.88540459e+18]]
 [01-21 11:21:10][INFO] i= 33 x: [[-2.96632117e+19 3.58735889e+19
 -3.33900111e+19 2.03739703e+19]]
 [01-21 11:21:10][INFO] i= 34 x: [[1.02653611e+20 -1.24175134e+20
 1.15573983e+20 -7.05053889e+19]]
 [01-21 11:21:10][INFO] i= 35 x: [[-3.55323099e+20 4.29743760e+20
 -3.99987745e+20 2.44049692e+20]]
 [01-21 11:21:10][INFO] i= 36 x: [[1.22971925e+21 -1.48745992e+21
 1.38443965e+21 -8.44609513e+20]]
 [01-21 11:21:10][INFO] i= 37 x: [[-4.25633922e+21 5.14798744e+21
 -4.79150793e+21 2.92340970e+21]]
 [01-21 11:21:10][INFO] i= 38 x: [[1.47310033e+22 -1.78180763e+22
 1.65840756e+22 -1.01177351e+22]]
 [01-21 11:21:10][INFO] i= 39 x: [[-5.09862274e+22 6.16682843e+22
 -5.73978180e+22 3.50191522e+22]]
 [01-21 11:21:10][INFO] i= 40 x: [[1.76463920e+23 -2.13441502e+23
 1.98659891e+23 -1.21201335e+23]]
 [01-21 11:21:10][INFO] i= 41 x: [[-6.10761285e+23 7.38727818e+23
 -6.87570678e+23 4.19492199e+23]]
 [01-21 11:21:10][INFO] i= 42 x: [[2.11386917e+24 -2.55680856e+24
 2.37974259e+24 -1.45187707e+24]]
 [01-21 11:21:10][INFO] i= 43 x: [[-7.31629373e+24 8.84924296e+24
 -8.23642397e+24 5.02508413e+24]]
 [01-21 11:21:10][INFO] i= 44 x: [[2.53220909e+25 -3.06279701e+25
 2.85069146e+25 -1.73920764e+25]]
 [01-21 11:21:10][INFO] i= 45 x: [[-8.76417992e+25 1.06005235e+26
 -9.86642283e+25 6.01954124e+25]]
 [01-21 11:21:10][INFO] i= 46 x: [[3.03333691e+26 -3.66892265e+26
 3.41484246e+26 -2.08339966e+26]]
 [01-21 11:21:10][INFO] i= 47 x: [[-1.04986076e+27 1.26983784e+27
 -1.18189961e+27 7.21080520e+26]]
 [01-21 11:21:10][INFO] i= 48 x: [[3.63363706e+27 -4.39500193e+27
 4.09063987e+27 -2.49570656e+27]]
 [01-21 11:21:10][INFO] i= 49 x: [[-1.25762817e+28 1.52113966e+28

-1.41579826e+28 8.63782283e+27]]



Solución encontrada: [-12576281663957068809743892480.000000,
15211396611257619653832212480.000000, -14157982615949674943979978752.000000,
8637822827778938123029839872.000000]

Caso 2: Posición inicial [1. 1. 1. 1.]

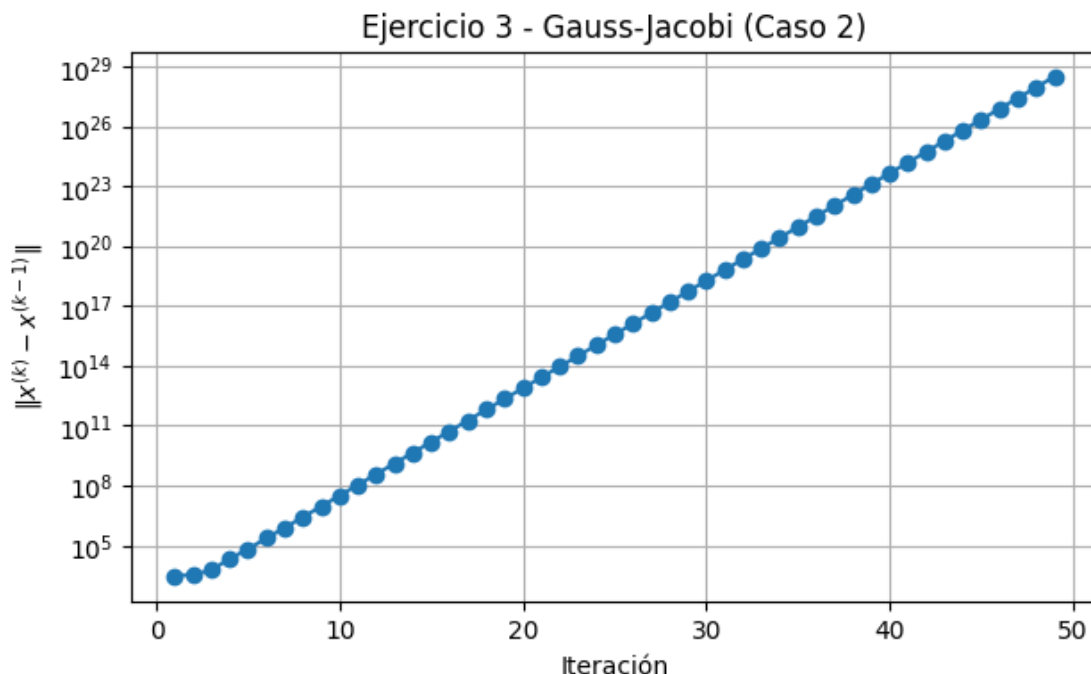
```
[01-21 11:21:10][INFO] i= 0 x: [1. 1. 1. 1.]
[01-21 11:21:10][INFO] i= 1 x: [[1359. 1129. 1350. 1824.]]
[01-21 11:21:10][INFO] i= 2 x: [[ -211.  -1633.  1585.   806.5]]
[01-21 11:21:10][INFO] i= 3 x: [[-3443.    84.5  945.5  5067. ]]
[01-21 11:21:10][INFO] i= 4 x: [[ -446.5 -16568.5  13387.5  2171. ]]
[01-21 11:21:10][INFO] i= 5 x: [[-41983.5  7558.  -12154.5  33371.5]]
[01-21 11:21:10][INFO] i= 6 x: [[ 33227.  -153122.5  126246.5  -15589.25]]
[01-21 11:21:10][INFO] i= 7 x: [[-404255.5  207371.25 -233815.75  294632. ]]
[01-21 11:21:10][INFO] i= 8 x: [[ 676362.75 -1520837.25  1311864.25
-426139.75]]
[01-21 11:21:10][INFO] i= 9 x: [[-4143205.75  3267776.25 -3298352.5  2939013.
]]
[01-21 11:21:10][INFO] i= 10 x: [[ 9865841.25 -16257467.25  14494550.75
-6549015.625]]
[01-21 11:21:10][INFO] i= 11 x: [[-45245208.75  44008568.875 -42536815.375
31635301.25 ]]
[01-21 11:21:10][INFO] i= 12 x: [[ 1.29083560e+08 -1.82686798e+08
1.66135638e+08 -8.72794360e+07]]
```


[01-21 11:21:10][INFO] i= 13 x: [[-5.14956713e+08 5.57058635e+08
 -5.28132003e+08 3.57099841e+08]]
 [01-21 11:21:10][INFO] i= 14 x: [[1.61332400e+09 -2.11438711e+09
 1.94407325e+09 -1.09965213e+09]]
 [01-21 11:21:10][INFO] i= 15 x: [[-6.00253225e+09 6.85635477e+09
 -6.44068589e+09 4.14361911e+09]]
 [01-21 11:21:10][INFO] i= 16 x: [[1.97377279e+10 -2.48740744e+10
 2.30050397e+10 -1.35048733e+10]]
 [01-21 11:21:10][INFO] i= 17 x: [[-7.08841525e+10 8.32573886e+10
 -7.78544021e+10 4.88136332e+10]]
 [01-21 11:21:10][INFO] i= 18 x: [[2.38966194e+11 -2.95179453e+11
 2.73839328e+11 -1.63813282e+11]]
 [01-21 11:21:10][INFO] i= 19 x: [[-8.42858108e+11 1.00424537e+12
 -9.36925122e+11 5.79688846e+11]]
 [01-21 11:21:10][INFO] i= 20 x: [[2.87809562e+12 -3.51884977e+12
 3.26965043e+12 -1.97483061e+12]]
 [01-21 11:21:10][INFO] i= 21 x: [[-1.00581506e+13 1.20722379e+13
 -1.12498716e+13 6.91309987e+12]]
 [01-21 11:21:10][INFO] i= 22 x: [[3.45719811e+13 -4.20473218e+13
 3.91016390e+13 -2.37332926e+13]]
 [01-21 11:21:10][INFO] i= 23 x: [[-1.20250600e+14 1.44873498e+14
 -1.34924577e+14 8.26218023e+13]]
 [01-21 11:21:10][INFO] i= 24 x: [[4.14722651e+14 -5.03040583e+14
 4.67996500e+14 -2.84772535e+14]]
 [01-21 11:21:10][INFO] i= 25 x: [[-1.43903358e+15 1.73703676e+15
 -1.61725842e+15 9.88559125e+14]]
 [01-21 11:21:10][INFO] i= 26 x: [[4.97155360e+15 -6.02196938e+15
 5.60366305e+15 -3.41418435e+15]]
 [01-21 11:21:10][INFO] i= 27 x: [[-1.72292955e+16 2.08177697e+16
 -1.93792609e+16 1.18347856e+16]]
 [01-21 11:21:10][INFO] i= 28 x: [[5.95762916e+16 -7.21129132e+16
 6.71111463e+16 -4.09162850e+16]]
 [01-21 11:21:10][INFO] i= 29 x: [[-2.06335206e+17 2.49436293e+17
 -2.32181781e+17 1.41724943e+17]]
 [01-21 11:21:10][INFO] i= 30 x: [[7.13799856e+17 -8.63691816e+17
 8.03831647e+17 -4.90245330e+17]]
 [01-21 11:21:10][INFO] i= 31 x: [[-2.47135511e+18 2.98836749e+18
 -2.78153686e+18 1.69745355e+18]]
 [01-21 11:21:10][INFO] i= 32 x: [[8.55144121e+18 -1.03452526e+19
 9.62853126e+18 -5.87331967e+18]]
 [01-21 11:21:10][INFO] i= 33 x: [[-2.96023151e+19 3.57999315e+19
 -3.33214547e+19 2.03321445e+19]]
 [01-21 11:21:10][INFO] i= 34 x: [[1.02442841e+20 -1.23920203e+20
 1.15336706e+20 -7.03606245e+19]]
 [01-21 11:21:10][INFO] i= 35 x: [[-3.54593616e+20 4.28861421e+20
 -3.99166510e+20 2.43548658e+20]]
 [01-21 11:21:10][INFO] i= 36 x: [[1.22719444e+21 -1.48440610e+21
 1.38159731e+21 -8.42875386e+20]]

```

[01-21 11:21:10][INFO] i= 37 x: [[-4.24760072e+21  5.13741791e+21
-4.78167037e+21  2.91740781e+21]]
[01-21 11:21:10][INFO] i= 38 x: [[ 1.47007586e+22 -1.77814945e+22
1.65500272e+22 -1.00969620e+22]]
[01-21 11:21:10][INFO] i= 39 x: [[-5.08815488e+22  6.15416719e+22
-5.72799738e+22  3.49472553e+22]]
[01-21 11:21:10][INFO] i= 40 x: [[ 1.76101620e+23 -2.13003289e+23
1.98252025e+23 -1.20952495e+23]]
[01-21 11:21:10][INFO] i= 41 x: [[-6.09507339e+23  7.37211129e+23
-6.86159023e+23  4.18630946e+23]]
[01-21 11:21:10][INFO] i= 42 x: [[ 2.10952917e+24 -2.55155920e+24
2.37485675e+24 -1.44889621e+24]]
[01-21 11:21:10][INFO] i= 43 x: [[-7.30127270e+24  8.83107454e+24
-8.21951375e+24  5.01476717e+24]]
[01-21 11:21:10][INFO] i= 44 x: [[ 2.52701020e+25 -3.05650880e+25
2.84483871e+25 -1.73563687e+25]]
[01-21 11:21:10][INFO] i= 45 x: [[-8.74618622e+25  1.05787595e+26
-9.84616608e+25  6.00718255e+25]]
[01-21 11:21:10][INFO] i= 46 x: [[ 3.02710917e+26 -3.66139000e+26
3.40783145e+26 -2.07912223e+26]]
[01-21 11:21:10][INFO] i= 47 x: [[-1.04770529e+27  1.26723073e+27
-1.17947306e+27  7.19600072e+26]]
[01-21 11:21:10][INFO] i= 48 x: [[ 3.62617684e+27 -4.38597856e+27
4.08224138e+27 -2.49058263e+27]]
[01-21 11:21:10][INFO] i= 49 x: [[-1.25504613e+28  1.51801661e+28
-1.41289149e+28  8.62008853e+27]]

```



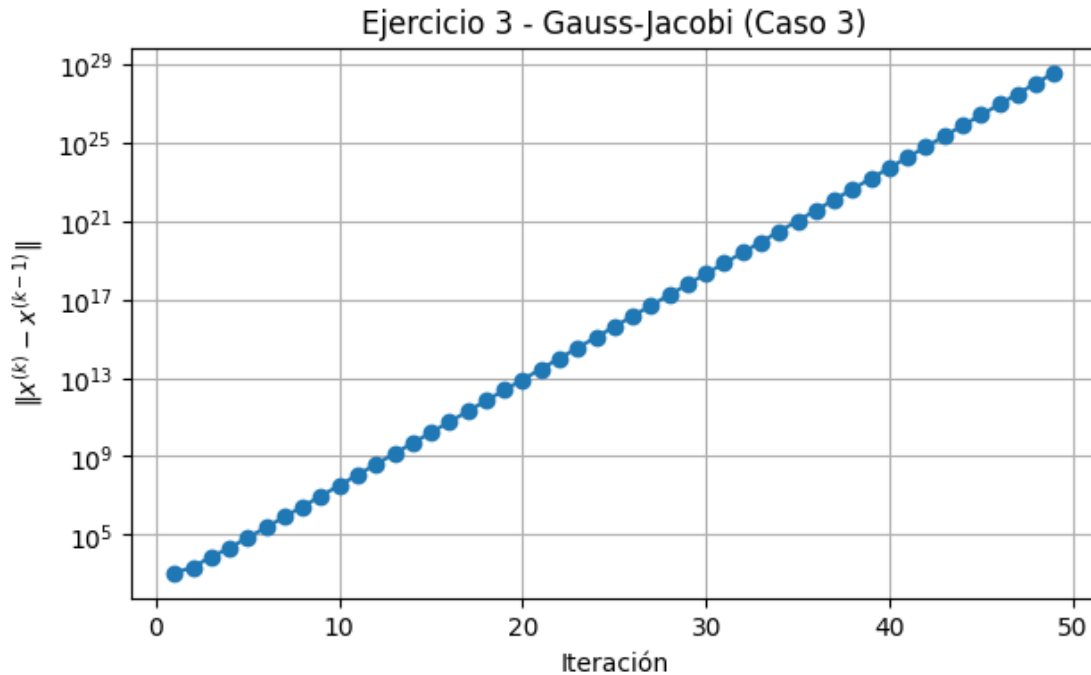
Solución encontrada: [-12550461328072733917634887680.000000,
15180166103750914500567498752.000000, -14128914875324018761670852608.000000,
8620088534147564163355901952.000000]

Caso 3: Posición inicial [1000. 1000. 1000. 1000.]

```
[01-21 11:21:10][INFO] i= 0 x: [1000. 1000. 1000. 1000.]
[01-21 11:21:10][INFO] i= 1 x: [[ 360.  130. 1350.  825.]]
[01-21 11:21:10][INFO] i= 2 x: [[-1210.  365. 1585. 2305.]]
[01-21 11:21:10][INFO] i= 3 x: [[-1445. -5410. 6440. 2070.]]
[01-21 11:21:10][INFO] i= 4 x: [[-16930.  -85.  900. 13160.]]
[01-21 11:21:10][INFO] i= 5 x: [[ -525. -54380. 48285. 2402.5]]
[01-21 11:21:10][INFO] i= 6 x: [[-149590.  41682.5 -49577.5 107537.5]]
[01-21 11:21:10][INFO] i= 7 x: [[ 142197.5 -520650. 449750. -85487.5]]
[01-21 11:21:10][INFO] i= 8 x: [[-1418790.  849540. -889182.5 1007675. ]]
[01-21 11:21:10][INFO] i= 9 x: [[ 2629265.  -5329867.5 4696145.
-1717076.25]]
[01-21 11:21:10][INFO] i= 10 x: [[-14720797.5 12477768.75 -12304123.75
10344698.75]]
[01-21 11:21:10][INFO] i= 11 x: [[ 37087376.25 -58057887.5 52265412.5
-24866890.  ]]
[01-21 11:21:10][INFO] i= 12 x: [[-1.62587352e+08 1.63954589e+08
-1.57098180e+08 1.13221362e+08]]
[01-21 11:21:10][INFO] i= 13 x: [[ 4.78152309e+08 -6.59348490e+08
6.02352006e+08 -3.24479148e+08]]
[01-21 11:21:10][INFO] i= 14 x: [[-1.86405114e+09 2.05394289e+09
-1.94013091e+09 1.29020056e+09]]
[01-21 11:21:10][INFO] i= 15 x: [[ 5.93420606e+09 -7.67478261e+09
7.07224709e+09 -4.05097796e+09]]
[01-21 11:21:10][INFO] i= 16 x: [[-2.18192754e+10 2.51593882e+10
-2.35941713e+10 1.50482993e+10]]
[01-21 11:21:10][INFO] i= 17 x: [[ 7.23477322e+10 -9.05583435e+10
8.38462396e+10 -4.95361661e+10]]
[01-21 11:21:10][INFO] i= 18 x: [[-2.58250821e+11 3.04802471e+11
-2.84789973e+11 1.77760637e+11]]
[01-21 11:21:10][INFO] i= 19 x: [[ 8.74382418e+11 -1.07632270e+12
9.99064752e+11 -5.99598691e+11]]
[01-21 11:21:10][INFO] i= 20 x: [[-3.07445221e+12 3.67224325e+12
-3.42468623e+12 2.11401643e+12]]
[01-21 11:21:10][INFO] i= 21 x: [[ 1.05216157e+13 -1.28411877e+13
1.19351641e+13 -7.22070798e+12]]
[01-21 11:21:10][INFO] i= 22 x: [[-3.67115159e+13 4.41189037e+13
-4.11051271e+13 2.52293636e+13]]
[01-21 11:21:10][INFO] i= 23 x: [[ 1.26329158e+14 -1.53504734e+14
1.42771299e+14 -8.67309192e+13]]
[01-21 11:21:10][INFO] i= 24 x: [[-4.39047332e+14 5.29293215e+14
-4.92893969e+14 3.01642751e+14]]
```

[01-21 11:21:10][INFO] i= 25 x: [[1.51508115e+15 -1.83686955e+15
 1.70903063e+15 -1.04038681e+15]]
 [01-21 11:21:10][INFO] i= 26 x: [[-5.25493081e+15 6.34527220e+15
 -5.90741867e+15 3.60981965e+15]]
 [01-21 11:21:10][INFO] i= 27 x: [[1.81601095e+16 -2.19918084e+16
 2.04649535e+16 -1.24716176e+16]]
 [01-21 11:21:10][INFO] i= 28 x: [[-6.29217154e+16 7.60399159e+16
 -7.07836451e+16 4.32201894e+16]]
 [01-21 11:21:10][INFO] i= 29 x: [[2.17607206e+17 -2.63365929e+17
 2.45103536e+17 -1.49451696e+17]]
 [01-21 11:21:10][INFO] i= 30 x: [[-7.53573001e+17 9.11065832e+17
 -8.48032037e+17 5.17600661e+17]]
 [01-21 11:21:10][INFO] i= 31 x: [[2.60712991e+18 -3.15440702e+18
 2.93581249e+18 -1.79061477e+18]]
 [01-21 11:21:10][INFO] i= 32 x: [[-9.02603201e+18 1.09147867e+19
 -1.01592816e+19 6.19951678e+18]]
 [01-21 11:21:10][INFO] i= 33 x: [[3.12333499e+19 -3.77838639e+19
 3.51663675e+19 -2.14518208e+19]]
 [01-21 11:21:10][INFO] i= 34 x: [[-1.08116599e+20 1.30755180e+20
 -1.21702385e+20 7.42589797e+19]]
 [01-21 11:21:10][INFO] i= 35 x: [[3.74159949e+20 -4.52595922e+20
 4.21247357e+20 -2.56983962e+20]]
 [01-21 11:21:10][INFO] i= 36 x: [[-1.29509064e+21 1.56635919e+21
 -1.45789978e+21 8.89517562e+20]]
 [01-21 11:21:10][INFO] i= 37 x: [[4.48215876e+21 -5.42154311e+21
 5.04605803e+21 -3.07848868e+21]]
 [01-21 11:21:10][INFO] i= 38 x: [[-1.55136592e+22 1.87636828e+22
 -1.74643493e+22 1.06553437e+22]]
 [01-21 11:21:10][INFO] i= 39 x: [[5.36923814e+22 -6.49440395e+22
 6.04463448e+22 -3.68776989e+22]]
 [01-21 11:21:10][INFO] i= 40 x: [[-1.85836729e+23 2.24771823e+23
 -2.09206501e+23 1.27639232e+23]]
 [01-21 11:21:10][INFO] i= 41 x: [[6.43184826e+23 -7.77960926e+23
 7.24084513e+23 -4.41760985e+23]]
 [01-21 11:21:10][INFO] i= 42 x: [[-2.22612995e+24 2.69255229e+24
 -2.50609156e+24 1.52898364e+24]]
 [01-21 11:21:10][INFO] i= 43 x: [[7.70473542e+24 -9.31917245e+24
 8.67379584e+24 -5.29187422e+24]]
 [01-21 11:21:10][INFO] i= 44 x: [[-2.66667641e+25 3.22541539e+25
 -3.00205175e+25 1.83156566e+25]]
 [01-21 11:21:10][INFO] i= 45 x: [[9.22951889e+25 -1.11634251e+26
 1.03903339e+26 -6.33914896e+25]]
 [01-21 11:21:10][INFO] i= 46 x: [[-3.19440929e+26 3.86372997e+26
 -3.59616119e+26 2.19403047e+26]]
 [01-21 11:21:10][INFO] i= 47 x: [[1.10560523e+27 -1.33726619e+27
 1.24465790e+27 -7.59367554e+26]]
 [01-21 11:21:10][INFO] i= 48 x: [[-3.82658199e+27 4.62836580e+27
 -4.30784421e+27 2.62822823e+27]]

[01-21 11:21:10] [INFO] i= 49 x: [[1.32440542e+28 -1.60191109e+28
1.49097580e+28 -9.09647080e+27]]



Solución encontrada: [13244054220375204262563545088.000000,
-1601911089544785906632337216.000000, 14909758009701392437053751296.000000,
-9096470803593889053931470848.000000]

```
[19]: import numpy as np
import matplotlib.pyplot as plt

def plot_error_convergence(trayectoria, titulo="Convergencia (Norma del_1
error)"):
    """
    Grafica la norma del cambio entre iteraciones:
        e_k = || x^(k) - x^(k-1) ||
    usando escala logarítmica en Y.
    """
    # Convertir cada x_k a vector 1D consistente
    X = []
    for xk in trayectoria:
        xk = np.array(xk, dtype=float)
        xk = xk.reshape(-1) # (n,1) o (n,) -> (n,)
        X.append(xk)
    X = np.array(X)
```

```

if len(X) < 2:
    print("No hay suficientes iteraciones para graficar.")
    return

errores = []
for k in range(1, len(X)):
    e = np.linalg.norm(X[k] - X[k-1])
    errores.append(e)

plt.figure(figsize=(7, 4))
plt.plot(range(1, len(errores) + 1), errores, marker='o')
plt.yscale('log')
plt.xlabel("Iteración")
plt.ylabel(r"$\|x^{\{k\}} - x^{\{k-1\}}\|$")
plt.title(titulo)
plt.grid(True)
plt.show()

# %% Ejercicio 3 - GAUSS-SEIDEL
A3 = np.array([
    [ 1, -1,  2,  0],
    [-1,  1, -1,  3],
    [ 2, -1,  1, -1],
    [ 0,  3, -1,  2]
], dtype=float)

b3 = np.array([1360, 1130, 1350, 3650], dtype=float)

initial_positions = [
    np.array([0, 0, 0, 0], dtype=float),
    np.array([1, 1, 1, 1], dtype=float),
    np.array([1000, 1000, 1000, 1000], dtype=float)
]

print("\n=== EJERCICIO 3 - MÉTODO DE GAUSS-SEIDEL ===")
for i, x0 in enumerate(initial_positions, 1):
    print(f"\nCaso {i}: Posición inicial {x0}")
    try:
        x_gs3, tray_gs3 = gauss_seidel(
            A=A3,
            b=b3,
            x0=x0,
            tol=1e-6,
            max_iter=200
        )

```

```

# "Trayectoria" para 4D: convergencia del error
plot_error_convergence(
    tray_gs3,
    titulo=f"Ejercicio 3 - Gauss-Seidel (Caso {i})"
)

print(
    "Solución encontrada: "
    f"{x_gs3[0,0]:.6f}, {x_gs3[1,0]:.6f}, "
    f"{x_gs3[2,0]:.6f}, {x_gs3[3,0]:.6f}"
)
except Exception as e:
    print(f";Error! El método divergió: {str(e)}")

```

=== EJERCICIO 3 - MÉTODO DE GAUSS-SEIDEL ===

Caso 1: Posición inicial [0. 0. 0. 0.]

```

[01-20 17:39:17][INFO] i= 0 x: [0. 0. 0. 0.]
[01-20 17:39:17][INFO] i= 1 x: [[ 1360.  2490.  1120. -1350.]]
[01-20 17:39:17][INFO] i= 2 x: [[ 1610.  7910.  4690. -7695.]]
[01-20 17:39:17][INFO] i= 3 x: [[ -110.  28795.  22670. -30032.5]]
[01-20 17:39:17][INFO] i= 4 x: [[-15185.  98712.5 100400. -96043.75]]
[01-20 17:39:17][INFO] i= 5 x: [[-100727.5  288933.75  395695.
-233728.125]]
[01-20 17:39:17][INFO] i= 6 x: [[-501096.25  596913.125 1366727.5
-210180.9375]]
[01-20 17:39:17][INFO] i= 7 x: [[-2135181.875  -136781.5625  3924751.25
2169372.96875]]
[01-20 17:39:17][INFO] i= 8 x: [[ -7984924.0625  -10567161.71875  7573409.375
19639272.265625]]
[01-20 17:39:17][INFO] i= 9 x: [[-2.57126205e+07 -7.70558979e+07 -5.99003469e+06
1.12590654e+08]]
[01-20 17:39:17][INFO] i= 10 x: [[-6.50744685e+07 -4.08835337e+08
-1.66094395e+08  5.30207632e+08]]
[01-20 17:39:17][INFO] i= 11 x: [[-7.66451864e+07 -1.83336135e+09
-1.14986199e+09  2.17511285e+09]]
[01-20 17:39:17][INFO] i= 12 x: [[ 4.66363998e+08 -7.20883542e+09
-5.96644922e+09  7.83003035e+09]]
[01-20 17:39:17][INFO] i= 13 x: [[ 4.72406437e+09 -2.47324748e+10
-2.63505718e+10  2.39234281e+10]]
[01-20 17:39:17][INFO] i= 14 x: [[ 2.79686702e+10 -7.01521846e+10
-1.02166096e+11  5.41452310e+10]]
[01-20 17:39:17][INFO] i= 15 x: [[ 1.34180008e+11 -1.30421779e+11
-3.44636563e+11  2.33143894e+10]]
[01-20 17:39:17][INFO] i= 16 x: [[ 5.58851348e+11  1.44271618e+11
-9.50116687e+11 -6.91465769e+11]]
[01-20 17:39:17][INFO] i= 17 x: [[ 2.04450499e+12  3.16878561e+12

```

-1.61169014e+12 -5.55902349e+12]]
 [01-20 17:39:17][INFO] i= 18 x: [[6.39216590e+12 2.14575462e+13
 3.11419094e+12 -3.06292239e+13]]
 [01-20 17:39:17][INFO] i= 19 x: [[1.52291643e+13 1.10231027e+14
 4.91434744e+13 -1.40774803e+14]]
 [01-20 17:39:17][INFO] i= 20 x: [[1.19440782e+13 4.83411962e+14
 3.18749003e+14 -5.65743442e+14]]
 [01-20 17:39:17][INFO] i= 21 x: [[-1.54086043e+14 1.86189329e+15
 1.60432193e+15 -1.99067896e+15]]
 [01-20 17:39:17][INFO] i= 22 x: [[-1.34675057e+15 6.22960825e+15
 6.93243043e+15 -5.87819716e+15]]
 [01-20 17:39:17][INFO] i= 23 x: [[-7.63525261e+15 1.69317693e+16
 2.63240774e+16 -1.22356152e+16]]
 [01-20 17:39:17][INFO] i= 24 x: [[-3.57163854e+16 2.73145377e+16
 8.65116933e+16 2.28404012e+15]]
 [01-20 17:39:17][INFO] i= 25 x: [[-1.45708849e+17 -6.60492760e+16
 2.27652462e+17 2.12900145e+17]]
 [01-20 17:39:17][INFO] i= 26 x: [[-5.21354200e+17 -9.32402173e+17
 3.23206372e+17 1.56020644e+18]]
 [01-20 17:39:17][INFO] i= 27 x: [[-1.57881492e+18 -5.93622788e+18
 -1.21839160e+18 8.29514602e+18]]
 [01-20 17:39:17][INFO] i= 28 x: [[-3.49944468e+18 -2.96032743e+19
 -1.43092390e+19 3.72502920e+19]]
 [01-20 17:39:17][INFO] i= 29 x: [[-9.84796409e+17 -1.27044911e+20
 -8.78250266e+19 1.46654854e+20]]
 [01-20 17:39:17][INFO] i= 30 x: [[4.86051417e+19 -4.79184446e+20
 -4.29739876e+20 5.03906731e+20]]
 [01-20 17:39:17][INFO] i= 31 x: [[3.80295306e+20 -1.56116476e+21
 -1.81784864e+21 1.43282282e+21]]
 [01-20 17:39:17][INFO] i= 32 x: [[2.07453252e+21 -4.04178459e+21
 -6.75802682e+21 2.68366348e+21]]
 [01-20 17:39:17][INFO] i= 33 x: [[9.47426904e+21 -5.33474822e+21
 -2.15996228e+22 -2.79768908e+21]]
 [01-20 17:39:17][INFO] i= 34 x: [[3.78644974e+22 2.46579418e+22
 -5.38687421e+22 -6.39212838e+22]]
 [01-20 17:39:17][INFO] i= 35 x: [[1.32395426e+23 2.70290535e+23
 -5.84216005e+22 -4.34646603e+23]]
 [01-20 17:39:17][INFO] i= 36 x: [[3.87133736e+23 1.63265195e+24
 4.23737870e+23 -2.23710898e+24]]
 [01-20 17:39:17][INFO] i= 37 x: [[7.85176207e+23 7.92024103e+24
 4.11277963e+24 -9.82397173e+24]]
 [01-20 17:39:17][INFO] i= 38 x: [[-3.05318234e+23 3.32793766e+25
 2.40660413e+25 -3.78860442e+25]]
 [01-20 17:39:17][INFO] i= 39 x: [[-1.48527061e+25 1.22871468e+26
 1.14690836e+26 -1.26961784e+26]]
 [01-20 17:39:17][INFO] i= 40 x: [[-1.06510204e+26 3.89065984e+26
 4.75124607e+26 -3.46036672e+26]]
 [01-20 17:39:17][INFO] i= 41 x: [[-5.61183231e+26 9.52051393e+26

1.72838118e+27 -5.63886498e+26]]
 [01-20 17:39:17][INFO] i= 42 x: [[-2.50471097e+27 9.15329705e+26
 5.36086515e+27 1.30743802e+27]]
 [01-20 17:39:17][INFO] i= 43 x: [[-9.80640060e+27 -8.36784950e+27
 1.25523897e+28 1.88279691e+28]]
 [01-20 17:39:17][INFO] i= 44 x: [[-3.34726289e+28 -7.74041465e+28
 8.36908042e+27 1.20290760e+29]]
 [01-20 17:39:17][INFO] i= 45 x: [[-9.41423074e+28 -4.46645507e+29
 -1.38070132e+29 6.00933194e+29]]
 [01-20 17:39:17][INFO] i= 46 x: [[-1.70505243e+29 -2.11137496e+30
 -1.16943128e+30 2.58234680e+30]]
 [01-20 17:39:17][INFO] i= 47 x: [[2.27487599e+29 -8.68898407e+30
 -6.56161247e+30 9.75266987e+30]]
 [01-20 17:39:17][INFO] i= 48 x: [[4.43424087e+30 -3.13853812e+31
 -3.05011931e+31 3.18274753e+31]]
 [01-20 17:39:17][INFO] i= 49 x: [[2.96170050e+31 -9.63666139e+31
 -1.23773149e+32 8.26633466e+31]]
 [01-20 17:39:17][INFO] i= 50 x: [[1.51179683e+32 -2.20583505e+32
 -4.40279525e+32 1.10735495e+32]]
 [01-20 17:39:17][INFO] i= 51 x: [[6.59975545e+32 -1.12510466e+32
 -1.32172606e+33 -4.92097331e+32]]
 [01-20 17:39:17][INFO] i= 52 x: [[2.53094166e+33 2.68550759e+33
 -2.86847305e+33 -5.46249791e+33]]
 [01-20 17:39:17][INFO] i= 53 x: [[8.42245369e+33 2.19414744e+34
 -3.65930931e+32 -3.30951770e+34]]
 [01-20 17:39:17][INFO] i= 54 x: [[2.26733362e+34 1.21592936e+35
 4.31510869e+34 -1.60813861e+35]]
 [01-20 17:39:17][INFO] i= 55 x: [[3.52907626e+34 5.60883433e+35
 3.29488046e+35 -6.76581126e+35]]
 [01-20 17:39:17][INFO] i= 56 x: [[-9.80926601e+34 2.26113876e+36
 1.78074296e+36 -2.50133667e+36]]
 [01-20 17:39:17][INFO] i= 57 x: [[-1.30034715e+36 7.98440581e+36
 8.08376344e+36 -7.93472699e+36]]
 [01-20 17:39:17][INFO] i= 58 x: [[-8.18312108e+36 2.37048233e+37
 3.21363385e+37 -1.94890657e+37]]
 [01-20 17:39:17][INFO] i= 59 x: [[-4.05678537e+37 5.00356820e+37
 1.11682324e+38 -1.92123612e+37]]
 [01-20 17:39:17][INFO] i= 60 x: [[-1.73328965e+38 -4.00955805e+36
 3.23436011e+38 1.67732343e+38]]
 [01-20 17:39:17][INFO] i= 61 x: [[-6.50881581e+38 -8.30642598e+38
 6.38852907e+38 1.56539035e+39]]
 [01-20 17:39:17][INFO] i= 62 x: [[-2.10834841e+39 -6.16566655e+39
 -3.83579382e+38 9.05671014e+39]]
 [01-20 17:39:17][INFO] i= 63 x: [[-5.39850779e+39 -3.29522176e+40
 -1.30984919e+40 4.28790804e+40]]
 [01-20 17:39:17][INFO] i= 64 x: [[-6.75523385e+39 -1.48490967e+41
 -9.21014189e+40 1.76685741e+41]]
 [01-20 17:39:17][INFO] i= 65 x: [[3.57118708e+40 -5.86446771e+41

-4.81184772e+41 6.39077771e+41]]
 [01-20 17:39:17][INFO] i= 66 x: [[3.75922772e+41 -2.02249531e+42
 -2.13526309e+42 1.96611143e+42]]
 [01-20 17:39:17][INFO] i= 67 x: [[2.24803086e+42 -5.78556651e+42
 -8.31551680e+42 4.52059136e+42]]
 [01-20 17:39:17][INFO] i= 68 x: [[1.08454671e+43 -1.10318238e+43
 -2.82021666e+43 2.44665237e+42]]
 [01-20 17:39:17][INFO] i= 69 x: [[4.53725094e+43 9.83038572e+42
 -7.84679808e+43 -5.39795690e+43]]
 [01-20 17:39:17][INFO] i= 70 x: [[1.66766347e+44 2.50237073e+44
 -1.37275190e+44 -4.43993205e+44]]
 [01-20 17:39:17][INFO] i= 71 x: [[5.24787454e+44 1.71949188e+45
 2.25923767e+44 -2.46627594e+45]]
 [01-20 17:39:17][INFO] i= 72 x: [[1.26764435e+45 8.89239592e+45
 3.89083129e+45 -1.13931782e+46]]
 [01-20 17:39:17][INFO] i= 73 x: [[1.11073334e+45 3.91810993e+46
 2.55664544e+46 -4.59884218e+46]]
 [01-20 17:39:17][INFO] i= 74 x: [[-1.19518095e+46 1.51579910e+47
 1.29495108e+47 -1.62622312e+47]]
 [01-20 17:39:17][INFO] i= 75 x: [[-1.07410305e+47 5.09951738e+47
 5.62150035e+47 -4.83852589e+47]]
 [01-20 17:39:17][INFO] i= 76 x: [[-6.14348333e+47 1.39935947e+48
 2.14420355e+48 -1.02693743e+48]]
 [01-20 17:39:17][INFO] i= 77 x: [[-2.88904762e+48 2.33596821e+48
 7.08712603e+48 3.96107005e+46]]
 [01-20 17:39:17][INFO] i= 78 x: [[-1.18382838e+49 -4.86998992e+48
 1.88461885e+49 1.67280791e+49]]
 [01-20 17:39:17][INFO] i= 79 x: [[-4.25623669e+49 -7.39004158e+49
 2.79523971e+49 1.24826822e+50]]
 [01-20 17:39:17][INFO] i= 80 x: [[-1.29805210e+50 -4.76333279e+50
 -9.18960373e+49 6.68551901e+50]]
 [01-20 17:39:17][INFO] i= 81 x: [[-2.92541205e+50 -2.39009294e+51
 -1.13645863e+51 3.01691010e+51]]
 [01-20 17:39:17][INFO] i= 82 x: [[-1.17175677e+50 -1.03043646e+52
 -7.05310315e+51 1.19299953e+52]]
 [01-20 17:39:17][INFO] i= 83 x: [[3.80184170e+51 -3.90412474e+52
 -3.47149355e+52 4.12044034e+52]]
 [01-20 17:39:17][INFO] i= 84 x: [[3.03886236e+52 -1.27939522e+53
 -1.47512366e+53 1.18153100e+53]]
 [01-20 17:39:17][INFO] i= 85 x: [[1.67085210e+53 -3.34886457e+53
 -5.50903776e+53 2.26877798e+53]]
 [01-20 17:39:17][INFO] i= 86 x: [[7.66921095e+53 -4.64616073e+53
 -1.77158047e+54 -1.88866123e+53]]
 [01-20 17:39:17][INFO] i= 87 x: [[3.07854486e+54 1.87356276e+54
 -4.47239308e+54 -5.04654068e+54]]
 [01-20 17:39:17][INFO] i= 88 x: [[1.08183489e+55 2.14855779e+55
 -5.19766063e+54 -3.48271972e+55]]
 [01-20 17:39:17][INFO] i= 89 x: [[3.18808992e+55 1.31164830e+56

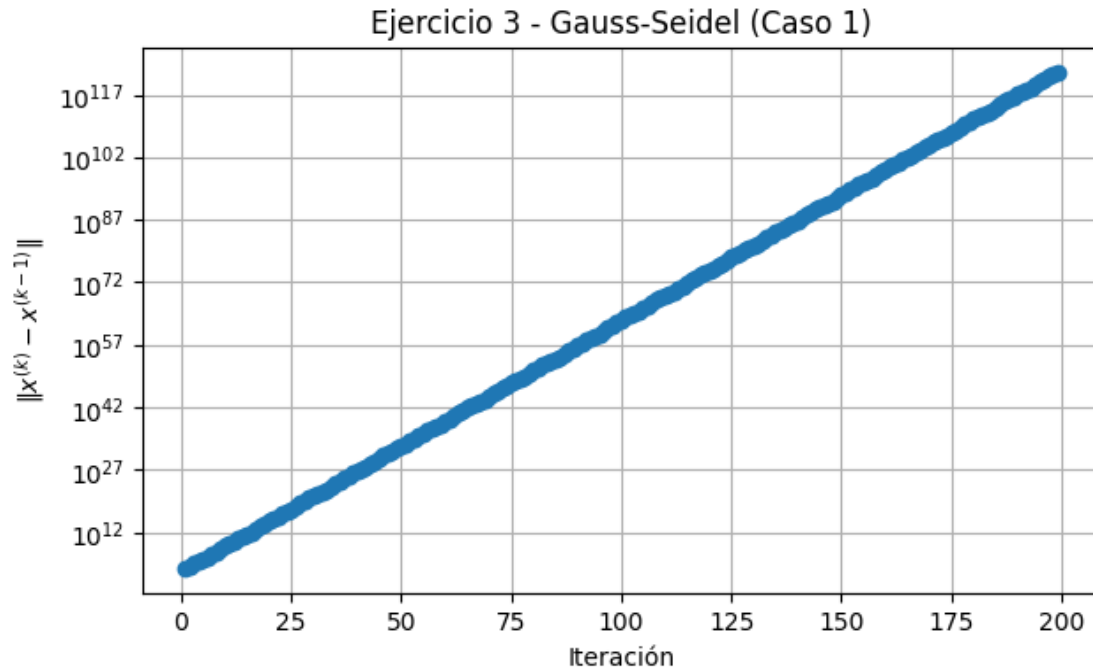
3.25758345e+55 -1.80459328e+56]]
 [01-20 17:39:17][INFO] i= 90 x: [[6.60131610e+55 6.39966979e+56
 3.27481329e+56 -7.96209804e+56]]
 [01-20 17:39:17][INFO] i= 91 x: [[-1.49956794e+55 2.70111506e+57
 1.93489662e+57 -3.08422428e+57]]
 [01-20 17:39:17][INFO] i= 92 x: [[-1.16867817e+57 1.00188913e+58
 9.27202335e+57 -1.03923253e+58]]
 [01-20 17:39:17][INFO] i= 93 x: [[-8.52515541e+57 3.19238437e+58
 3.85818293e+58 -2.85948510e+58]]
 [01-20 17:39:17][INFO] i= 94 x: [[-4.52398148e+58 7.91265673e+58
 1.41011346e+59 -4.81841779e+58]]
 [01-20 17:39:17][INFO] i= 95 x: [[-2.02896125e+59 8.26677551e+58
 4.40275827e+59 9.61362808e+58]]
 [01-20 17:39:17][INFO] i= 96 x: [[-7.97883898e+59 -6.46016914e+59
 1.04588716e+60 1.49196895e+60]]
 [01-20 17:39:17][INFO] i= 97 x: [[-2.73779124e+60 -6.16781094e+60
 7.99740499e+59 9.65158665e+60]]
 [01-20 17:39:17][INFO] i= 98 x: [[-7.76729193e+60 -3.59223114e+61
 -1.07361409e+61 4.85153967e+61]]
 [01-20 17:39:17][INFO] i= 99 x: [[-1.44500296e+61 -1.70732360e+62
 -9.33169045e+61 2.09440088e+62]]
 [01-20 17:39:17][INFO] i= 100 x: [[1.59014486e+61 -7.05735721e+62
 -5.28098530e+62 7.94554317e+62]]
 [01-20 17:39:17][INFO] i= 101 x: [[3.50461339e+62 -2.56130014e+63
 -2.46766850e+63 2.60811596e+63]]
 [01-20 17:39:17][INFO] i= 102 x: [[2.37403686e+63 -7.91797953e+63
 -1.00579373e+64 6.84800064e+63]]
 [01-20 17:39:17][INFO] i= 103 x: [[1.21978951e+64 -1.84040442e+64
 -3.59518336e+64 9.63014943e+63]]
 [01-20 17:39:17][INFO] i= 104 x: [[5.34996231e+64 -1.13426588e+64
 -1.08711756e+65 -3.73418895e+64]]
 [01-20 17:39:17][INFO] i= 105 x: [[2.06080852e+65 2.09394765e+65
 -2.40108829e+65 -4.34146563e+65]]
 [01-20 17:39:17][INFO] i= 106 x: [[6.89612423e+65 1.75194328e+66
 -6.14281269e+64 -2.65862899e+66]]
 [01-20 17:39:17][INFO] i= 107 x: [[1.87479954e+66 9.78925837e+66
 3.38103031e+66 -1.29933724e+67]]
 [01-20 17:39:17][INFO] i= 108 x: [[3.02719775e+66 4.53883452e+67
 2.63405774e+67 -5.49122292e+67]]
 [01-20 17:39:17][INFO] i= 109 x: [[-7.29280946e+66 1.83784455e+68
 1.43457845e+68 -2.03947761e+68]]
 [01-20 17:39:17][INFO] i= 110 x: [[-1.03131235e+68 6.52169892e+68
 6.54484601e+68 -6.51012537e+68]]
 [01-20 17:39:17][INFO] i= 111 x: [[-6.56799311e+68 1.95072290e+69
 2.61330899e+69 -1.61942986e+69]]
 [01-20 17:39:17][INFO] i= 112 x: [[-3.27589507e+69 4.19570350e+69
 9.12806378e+69 -1.72952336e+69]]
 [01-20 17:39:17][INFO] i= 113 x: [[-1.40604241e+70 2.56209798e+68

2.66475346e+70 1.29394526e+70]]
 [01-20 17:39:17][INFO] i= 114 x: [[-5.30388593e+70 -6.52096825e+70
 5.38074887e+70 1.24718268e+71]]
 [01-20 17:39:17][INFO] i= 115 x: [[-1.72824660e+71 -4.93171976e+71
 -2.28043876e+70 7.28355770e+71]]
 [01-20 17:39:17][INFO] i= 116 x: [[-4.47563200e+71 -2.65543490e+72
 -1.03195273e+72 3.46717598e+72]]
 [01-20 17:39:17][INFO] i= 117 x: [[-5.91529444e+71 -1.20250101e+73
 -7.37477525e+72 1.43501276e+73]]
 [01-20 17:39:17][INFO] i= 118 x: [[2.72454038e+72 -4.77006175e+73
 -3.87995707e+73 5.21511409e+73]]
 [01-20 17:39:17][INFO] i= 119 x: [[2.98985239e+73 -1.65354470e+74
 -1.73000376e+74 1.61531516e+74]]
 [01-20 17:39:17][INFO] i= 120 x: [[1.80646283e+74 -4.76948641e+74
 -6.76709692e+74 3.77068116e+74]]
 [01-20 17:39:17][INFO] i= 121 x: [[8.76470743e+74 -9.31443297e+74
 -2.30731667e+75 2.43506612e+74]]
 [01-20 17:39:17][INFO] i= 122 x: [[3.68319004e+75 6.45353532e+74
 -6.47751993e+75 -4.20679026e+75]]
 [01-20 17:39:17][INFO] i= 123 x: [[1.36003934e+76 1.97432442e+76
 -1.16643328e+76 -3.54470328e+76]]
 [01-20 17:39:17][INFO] i= 124 x: [[4.30719098e+76 1.37748675e+77
 1.61578229e+76 -1.98544102e+77]]
 [01-20 17:39:17][INFO] i= 125 x: [[1.05433030e+77 7.17223157e+77
 3.07812997e+77 -9.21928237e+77]]
 [01-20 17:39:17][INFO] i= 126 x: [[1.01597164e+77 3.17519487e+78
 2.05007231e+78 -3.73775616e+78]]
 [01-20 17:39:17][INFO] i= 127 x: [[-9.24949742e+77 1.23383910e+79
 1.04505344e+79 -1.32823194e+79]]
 [01-20 17:39:17][INFO] i= 128 x: [[-8.56267769e+78 4.17348148e+79
 4.55778508e+79 -3.98132968e+79]]
 [01-20 17:39:17][INFO] i= 129 x: [[-4.94208868e+79 1.15596854e+80
 1.74625331e+80 -8.60826159e+79]]
 [01-20 17:39:17][INFO] i= 130 x: [[-2.33653808e+80 1.99219371e+80
 5.80444371e+80 -8.60687093e+78]]
 [01-20 17:39:17][INFO] i= 131 x: [[-9.61669371e+80 -3.55404387e+80
 1.55932748e+81 1.31277032e+81]]
 [01-20 17:39:17][INFO] i= 132 x: [[-3.47405935e+81 -5.85304284e+81
 2.40784619e+81 9.98348735e+81]]
 [01-20 17:39:17][INFO] i= 133 x: [[-1.06687352e+82 -3.82113511e+82
 -6.89039329e+81 5.38718300e+82]]
 [01-20 17:39:17][INFO] i= 134 x: [[-2.44305645e+82 -1.92936448e+83
 -9.02034888e+82 2.44302927e+83]]
 [01-20 17:39:17][INFO] i= 135 x: [[-1.25294702e+82 -8.35641741e+83
 -5.66279873e+83 9.70322675e+83]]
 [01-20 17:39:17][INFO] i= 136 x: [[2.96918005e+83 -3.18032989e+84
 -2.80384323e+84 3.36857322e+84]]
 [01-20 17:39:17][INFO] i= 137 x: [[2.42735656e+84 -1.04822063e+85

-1.19683462e+85 9.73913639e+84]]
 [01-20 17:39:17][INFO] i= 138 x: [[1.34544861e+85 -2.77312693e+85
 -4.49011052e+85 1.91463513e+85]]
 [01-20 17:39:17][INFO] i= 139 x: [[6.20709411e+85 -4.02692180e+85
 -1.45264749e+86 -1.22285474e+85]]
 [01-20 17:39:17][INFO] i= 140 x: [[2.50260280e+86 1.41681173e+86
 -3.71067934e+86 -3.98055726e+86]]
 [01-20 17:39:17][INFO] i= 141 x: [[8.83817040e+86 1.70691629e+87
 -4.58773521e+86 -2.78976119e+87]]
 [01-20 17:39:17][INFO] i= 142 x: [[2.62446333e+87 1.05349734e+88
 2.49628553e+87 -1.45543173e+88]]
 [01-20 17:39:17][INFO] i= 143 x: [[5.54240232e+87 5.17016397e+88
 2.60625178e+88 -6.45212007e+88]]
 [01-20 17:39:17][INFO] i= 144 x: [[-4.23395877e+86 2.19202724e+89
 1.55528315e+89 -2.51039929e+89]]
 [01-20 17:39:17][INFO] i= 145 x: [[-9.18539061e+88 8.16794195e+89
 7.49462078e+89 -8.50460253e+89]]
 [01-20 17:39:17][INFO] i= 146 x: [[-6.82129962e+89 2.61871287e+90
 3.13251255e+90 -2.36181304e+90]]
 [01-20 17:39:17][INFO] i= 147 x: [[-3.64631222e+90 6.57163944e+90
 1.15024508e+91 -4.10623375e+90]]
 [01-20 17:39:17][INFO] i= 148 x: [[-1.64332622e+91 7.38788984e+90
 3.61481806e+91 6.99225552e+90]]
 [01-20 17:39:17][INFO] i= 149 x: [[-6.49084713e+91 -4.97370573e+91
 8.70721408e+91 1.18141656e+92]]
 [01-20 17:39:17][INFO] i= 150 x: [[-2.23881339e+92 -4.91234167e+92
 7.46701671e+91 7.74186334e+92]]
 [01-20 17:39:17][INFO] i= 151 x: [[-6.40574501e+92 -2.88846334e+93
 -8.33128000e+92 3.91613101e+93]]
 [01-20 17:39:17][INFO] i= 152 x: [[-1.22220734e+93 -1.38037284e+94
 -7.44318267e+93 1.69840012e+94]]
 [01-20 17:39:17][INFO] i= 153 x: [[1.08263700e+93 -5.73125493e+94
 -4.24938221e+94 6.47219129e+94]]
 [01-20 17:39:17][INFO] i= 154 x: [[2.76750949e+94 -2.08984466e+95
 -1.99612743e+95 2.13670327e+95]]
 [01-20 17:39:17][INFO] i= 155 x: [[1.90241019e+95 -6.50382705e+95
 -8.17194417e+95 5.66976849e+95]]
 [01-20 17:39:17][INFO] i= 156 x: [[9.84006128e+95 -1.53411884e+96
 -2.93515424e+96 8.33601133e+95]]
 [01-20 17:39:17][INFO] i= 157 x: [[4.33618965e+96 -1.09976799e+96
 -8.93854616e+96 -2.81962109e+96]]
 [01-20 17:39:17][INFO] i= 158 x: [[1.67773243e+97 1.62976414e+97
 -2.00766283e+97 -3.44847763e+97]]
 [01-20 17:39:17][INFO] i= 159 x: [[5.64508981e+97 1.39828599e+98
 -7.55797371e+96 -2.13521885e+98]]
 [01-20 17:39:17][INFO] i= 160 x: [[1.54944546e+98 7.87952227e+98
 2.64541250e+98 -1.04965772e+99]]
 [01-20 17:39:17][INFO] i= 161 x: [[2.58869727e+98 3.67238412e+99

2.10498695e+99 -4.45608271e+99]]
 [01-20 17:39:17][INFO] i= 162 x: [[-5.37589784e+098 1.49356453e+100
 1.15547422e+100 -1.66260969e+100]]
 [01-20 17:39:17][INFO] i= 163 x: [[-8.17383902e+099 5.32591938e+100
 5.29807749e+100 -5.33984032e+100]]
 [01-20 17:39:17][INFO] i= 164 x: [[-5.27023561e+100 1.60473628e+101
 2.12479937e+101 -1.34470474e+101]]
 [01-20 17:39:17][INFO] i= 165 x: [[-2.64486246e+101 3.51405113e+101
 7.45907131e+101 -1.54154103e+101]]
 [01-20 17:39:17][INFO] i= 166 x: [[-1.14040915e+102 6.79602906e+100
 2.19462449e+102 9.95371808e+101]]
 [01-20 17:39:17][INFO] i= 167 x: [[-4.32128869e+102 -5.11277962e+102
 4.52516956e+102 9.93175421e+102]]
 [01-20 17:39:17][INFO] i= 168 x: [[-1.41631187e+103 -3.94332118e+103
 -1.17522013e+102 5.85622077e+103]]
 [01-20 17:39:17][INFO] i= 169 x: [[-3.70827716e+103 -2.13944615e+104
 -8.12168639e+103 2.80308490e+104]]
 [01-20 17:39:17][INFO] i= 170 x: [[-5.15108869e+103 -9.73653221e+104
 -5.90322957e+104 1.16531835e+105]]
 [01-20 17:39:17][INFO] i= 171 x: [[2.06992693e+104 -3.87928532e+105
 -3.12795236e+105 4.25495180e+105]]
 [01-20 17:39:17][INFO] i= 172 x: [[2.37661939e+105 -1.35161884e+106
 -1.40144754e+106 1.32670449e+106]]
 [01-20 17:39:17][INFO] i= 173 x: [[1.45127623e+106 -3.93028477e+106
 -5.50613275e+106 3.14236078e+106]]
 [01-20 17:39:17][INFO] i= 174 x: [[7.08198072e+106 -7.85123437e+106
 -1.88728350e+107 2.34043404e+106]]
 [01-20 17:39:17][INFO] i= 175 x: [[2.98944357e+107 4.00029855e+106
 -5.34481388e+107 -3.27245172e+107]]
 [01-20 17:39:17][INFO] i= 176 x: [[1.10896576e+108 1.55621989e+108
 -9.88956805e+107 -2.82880824e+108]]
 [01-20 17:39:17][INFO] i= 177 x: [[3.53413350e+108 1.10316014e+109
 1.13452617e+108 -1.59801390e+109]]
 [01-20 17:39:17][INFO] i= 178 x: [[8.76254907e+108 5.78374923e+109
 2.43322552e+109 -7.45901109e+109]]
 [01-20 17:39:17][INFO] i= 179 x: [[9.17298201e+108 2.57275570e+110
 1.64339495e+110 -3.03743607e+110]]
 [01-20 17:39:17][INFO] i= 180 x: [[-7.14034201e+109 1.00416690e+111
 8.43230130e+110 -1.08463528e+111]]
 [01-20 17:39:17][INFO] i= 181 x: [[-6.82293363e+110 3.41484261e+111
 3.69479405e+111 -3.27486689e+111]]
 [01-20 17:39:17][INFO] i= 182 x: [[-3.97474550e+111 9.54464922e+111
 1.42192733e+112 -7.20733716e+111]]
 [01-20 17:39:17][INFO] i= 183 x: [[-1.88938974e+112 1.69473874e+112
 4.75278451e+112 -1.65715853e+111]]
 [01-20 17:39:17][INFO] i= 184 x: [[-7.81083028e+112 -2.56089821e+112
 1.28950465e+113 1.02888706e+113]]
 [01-20 17:39:17][INFO] i= 185 x: [[-2.83509912e+113 -4.63225564e+113

2.06682966e+113 7.98179829e+113]]
 [01-20 17:39:17][INFO] i= 186 x: [[-8.76591495e+113 -3.06444802e+114
 -5.13085197e+113 4.34012943e+114]]
 [01-20 17:39:17][INFO] i= 187 x: [[-2.03827762e+114 -1.55717511e+115
 -7.15506643e+114 1.97800934e+115]]
 [01-20 17:39:17][INFO] i= 188 x: [[-1.26161825e+114 -6.77569650e+115
 -4.54536351e+115 7.89086299e+115]]
 [01-20 17:39:17][INFO] i= 189 x: [[2.31503051e+115 -2.59029220e+116
 -2.26421200e+116 2.75333230e+116]]
 [01-20 17:39:17][INFO] i= 190 x: [[1.93813180e+116 -8.58607708e+116
 -9.70900840e+116 8.02461143e+116]]
 [01-20 17:39:17][INFO] i= 191 x: [[1.08319397e+117 -2.29509030e+117
 -3.65901710e+117 1.61312690e+117]]
 [01-20 17:39:17][INFO] i= 192 x: [[5.02294389e+117 -3.47545390e+117
 -1.19082148e+118 -7.40926550e+116]]
 [01-20 17:39:17][INFO] i= 193 x: [[2.03409757e+118 1.06555405e+118
 -3.07673374e+118 -3.13669795e+118]]
 [01-20 17:39:17][INFO] i= 194 x: [[7.21902153e+118 1.35523816e+119
 -4.02235936e+118 -2.23397521e+119]]
 [01-20 17:39:17][INFO] i= 195 x: [[2.15971004e+119 8.45939974e+119
 1.90600445e+119 -1.17360974e+120]]
 [01-20 17:39:17][INFO] i= 196 x: [[4.64739083e+119 4.17616874e+120
 2.07308084e+120 -5.22771270e+120]]
 [01-20 17:39:17][INFO] i= 197 x: [[3.00070658e+118 1.77862260e+121
 1.24984992e+121 -2.04300894e+121]]
 [01-20 17:39:17][INFO] i= 198 x: [[-7.21077234e+120 6.65779951e+121
 6.05694503e+121 -6.95822674e+121]]
 [01-20 17:39:17][INFO] i= 199 x: [[-5.45609056e+121 2.14755347e+122
 2.54294891e+122 -1.94985575e+122]]



Solución encontrada: [-

54560905598530354936468235304488319502330379212816214796420001619944122473115272
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 898925919462230489133318981640626634752.000000]

Caso 2: Posición inicial [1. 1. 1. 1.]

[01-20 17:39:17][INFO] i= 0 x: [1. 1. 1. 1.]

[01-20 17:39:17][INFO] i= 1 x: [[1359. 2487. 1120. -1345.5]]

[01-20 17:39:17][INFO] i= 2 x: [[1607. 7893.5 4684. -7673.25]]

[01-20 17:39:17][INFO] i= 3 x: [[-114.5 28719.25 22625. -29941.375]]

[01-20 17:39:17][INFO] i= 4 x: [[-15170.75 98408.375 100158.5
 -95708.3125]]

[01-20 17:39:17][INFO] i= 5 x: [[-100548.625 287864.8125 394603.75
 -232670.34375]]

[01-20 17:39:17][INFO] i= 6 x: [[-499982.6875 593762.09375 1362407.125
 -207614.578125]]

[01-20 17:39:17][INFO] i= 7 x: [[-2129692.15625 -143311.296875 3909808.4375
 2171696.1640625]]

[01-20 17:39:17][INFO] i= 8 x: [[-7961568.171875 -10565718.2265625
 7530464.28125

19615634.48046875]]

[01-20 17:39:17][INFO] i= 9 x: [[-2.56252868e+07 -7.69405959e+07 -6.07303789e+06
1.12376200e+08]]

[01-20 17:39:17][INFO] i= 10 x: [[-6.47931602e+07 -4.07993668e+08
-1.66029798e+08 5.28977428e+08]]

[01-20 17:39:17][INFO] i= 11 x: [[-7.59327126e+07 -1.82889366e+09
-1.14804946e+09 2.16931759e+09]]

[01-20 17:39:17][INFO] i= 12 x: [[4.67206618e+08 -7.18879449e+09
-5.95388878e+09 7.80624917e+09]]

[01-20 17:39:17][INFO] i= 13 x: [[4.71898444e+09 -2.46536507e+10
-2.62853691e+10 2.38377934e+10]]

[01-20 17:39:17][INFO] i= 14 x: [[2.79170888e+10 -6.98816592e+10
-1.01878042e+11 5.38834696e+10]]

[01-20 17:39:17][INFO] i= 15 x: [[1.33874426e+11 -1.29654024e+11
-3.43519405e+11 2.27213345e+10]]

[01-20 17:39:17][INFO] i= 16 x: [[5.57384788e+11 1.45701381e+11
-9.46346860e+11 -6.91725499e+11]]

[01-20 17:39:17][INFO] i= 17 x: [[2.03839510e+12 3.16722474e+12
-1.60129096e+12 -5.55148259e+12]]

[01-20 17:39:17][INFO] i= 18 x: [[6.36980666e+12 2.14229635e+13
3.13186756e+12 -3.05685114e+13]]

[01-20 17:39:17][INFO] i= 19 x: [[1.51592284e+13 1.09996630e+14
4.91096621e+13 -1.40440114e+14]]

[01-20 17:39:17][INFO] i= 20 x: [[1.17773061e+13 4.82207311e+14
3.18212585e+14 -5.64204674e+14]]

[01-20 17:39:17][INFO] i= 21 x: [[-1.54217858e+14 1.85660875e+15
1.60083979e+15 -1.98449323e+15]]

[01-20 17:39:17][INFO] i= 22 x: [[-1.34507083e+15 6.20924864e+15
6.91489708e+15 -5.85642442e+15]]

[01-20 17:39:17][INFO] i= 23 x: [[-7.62054552e+15 1.68636248e+16
2.62482914e+16 -1.21712915e+16]]

[01-20 17:39:17][INFO] i= 24 x: [[-3.56329581e+16 2.71292080e+16
8.62238326e+16 2.41810432e+15]]

[01-20 17:39:17][INFO] i= 25 x: [[-1.45318457e+17 -6.63489376e+16
2.26706081e+17 2.12876447e+17]]

[01-20 17:39:17][INFO] i= 26 x: [[-5.19761100e+17 -9.31684359e+17
3.20714287e+17 1.55788368e+18]]

[01-20 17:39:17][INFO] i= 27 x: [[-1.57311293e+18 -5.92604969e+18
-1.22194014e+18 8.27810447e+18]]

[01-20 17:39:17][INFO] i= 28 x: [[-3.48216940e+18 -2.95384230e+19
-1.42959797e+19 3.71596446e+19]]

[01-20 17:39:17][INFO] i= 29 x: [[-9.46463601e+17 -1.26721377e+20
-8.76688053e+19 1.46247663e+20]]

[01-20 17:39:17][INFO] i= 30 x: [[4.86162335e+19 -4.77795561e+20
-4.28780365e+20 5.02303159e+20]]

[01-20 17:39:17][INFO] i= 31 x: [[3.79765169e+20 -1.55592467e+21
-1.81315185e+21 1.42731108e+21]]

[01-20 17:39:17][INFO] i= 32 x: [[2.07037903e+21 -4.02470607e+21
-6.73815304e+21 2.66798258e+21]]

[01-20 17:39:17][INFO] i= 33 x: [[9.45160002e+21 -5.29050077e+21
 -2.15257182e+22 -2.82710796e+21]]
 [01-20 17:39:17][INFO] i= 34 x: [[3.77609357e+22 2.47165413e+22
 -5.36324380e+22 -6.38910310e+22]]
 [01-20 17:39:17][INFO] i= 35 x: [[1.31981417e+23 2.70022072e+23
 -5.78317933e+22 -4.33949005e+23]]
 [01-20 17:39:17][INFO] i= 36 x: [[3.85685659e+23 1.62970088e+24
 4.24380558e+23 -2.23236104e+24]]
 [01-20 17:39:17][INFO] i= 37 x: [[7.80939765e+23 7.90240345e+24
 4.10816288e+24 -9.79952374e+24]]
 [01-20 17:39:17][INFO] i= 38 x: [[-3.13922305e+23 3.31928118e+25
 2.40211327e+25 -3.77786513e+25]]
 [01-20 17:39:17][INFO] i= 39 x: [[-1.48494535e+25 1.22507633e+26
 1.14427889e+26 -1.26547505e+26]]
 [01-20 17:39:17][INFO] i= 40 x: [[-1.06348145e+26 3.87722260e+26
 4.73871044e+26 -3.44647868e+26]]
 [01-20 17:39:17][INFO] i= 41 x: [[-5.60019828e+26 9.47794821e+26
 1.72318661e+27 -5.60098927e+26]]
 [01-20 17:39:17][INFO] i= 42 x: [[-2.49857840e+27 9.04904993e+26
 5.34196286e+27 1.31362394e+27]]
 [01-20 17:39:17][INFO] i= 43 x: [[-9.77902073e+27 -8.37792968e+27
 1.24937357e+28 1.88137624e+28]]
 [01-20 17:39:17][INFO] i= 44 x: [[-3.33654011e+28 -7.73129525e+28
 8.23161204e+27 1.20085235e+29]]
 [01-20 17:39:17][INFO] i= 45 x: [[-9.37761766e+28 -4.45800269e+29
 -1.38162681e+29 5.99619063e+29]]
 [01-20 17:39:17][INFO] i= 46 x: [[-1.69474907e+29 -2.10649478e+30
 -1.16792590e+30 2.57577922e+30]]
 [01-20 17:39:17][INFO] i= 47 x: [[2.29357023e+29 -8.66590652e+30
 -6.54884135e+30 9.72443911e+30]]
 [01-20 17:39:17][INFO] i= 48 x: [[4.43177618e+30 -3.12903825e+31
 -3.04294958e+31 3.17208259e+31]]
 [01-20 17:39:17][INFO] i= 49 x: [[2.95686090e+31 -9.60233643e+31
 -1.23439757e+32 8.23151683e+31]]
 [01-20 17:39:17][INFO] i= 50 x: [[1.50856149e+32 -2.19529113e+32
 -4.38926242e+32 1.09830548e+32]]
 [01-20 17:39:17][INFO] i= 51 x: [[6.58323371e+32 -1.10094515e+32
 -1.31691071e+33 -4.93313582e+32]]
 [01-20 17:39:17][INFO] i= 52 x: [[2.52372690e+33 2.68675694e+33
 -2.85401045e+33 -5.45714063e+33]]
 [01-20 17:39:17][INFO] i= 53 x: [[8.39477783e+33 2.19121893e+34
 -3.34507020e+32 -3.30355374e+34]]
 [01-20 17:39:17][INFO] i= 54 x: [[2.25812033e+34 1.21353309e+35
 4.31553645e+34 -1.60452281e+35]]
 [01-20 17:39:17][INFO] i= 55 x: [[3.50425795e+34 5.59554786e+35
 3.29017346e+35 -6.74823506e+35]]
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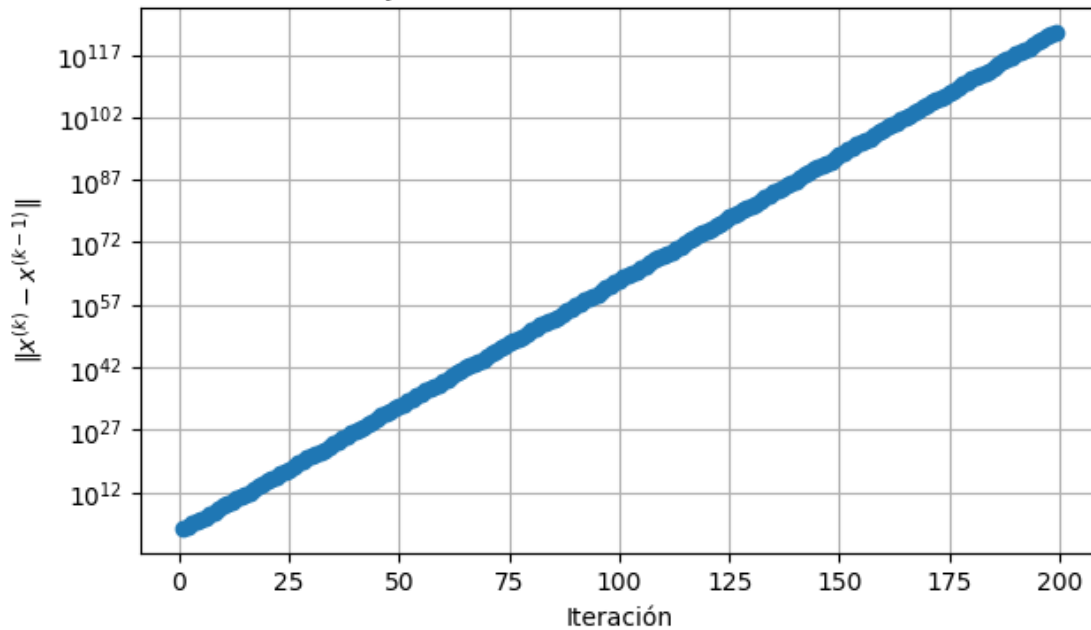
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 [01-20 17:39:17][INFO] i= 192 x: [[5.01110799e+117 -3.45033116e+117
 -1.18682017e+118 -7.58604130e+116]]
 [01-20 17:39:17][INFO] i= 193 x: [[2.02860723e+118 1.06936830e+118
 -3.06370658e+118 -3.13590574e+118]]
 [01-20 17:39:17][INFO] i= 194 x: [[7.19678146e+118 1.35407921e+119
 -3.98867657e+118 -2.23055264e+119]]
 [01-20 17:39:17][INFO] i= 195 x: [[2.15181452e+119 8.44460479e+119
 1.91042310e+119 -1.17116956e+120]]
 [01-20 17:39:17][INFO] i= 196 x: [[4.62375858e+119 4.16692686e+120
 2.07100558e+120 -5.21488750e+120]]
 [01-20 17:39:17][INFO] i= 197 x: [[2.49157010e+118 1.77405838e+121
 1.24758649e+121 -2.03729432e+121]]
 [01-20 17:39:17][INFO] i= 198 x: [[-7.21114597e+120 6.63835486e+121
 6.04328973e+121 -6.93588742e+121]]
 [01-20 17:39:17][INFO] i= 199 x: [[-5.44822460e+121 2.14027274e+122
 2.53632892e+122 -1.94224465e+122]]

Ejercicio 3 - Gauss-Seidel (Caso 2)



Solución encontrada: [-

54482246024235321310025747035291846839580800881084953023988054490047302865794173
646049461567016673076566110839178679812096.000000, 21402727392759171400195823772
17433032642017533968442207844274577872276471852859521436160831671921371890403252
53069728120832.000000, 253632891759213570774141843201510705584968786448428525600
034584138720796395479956566700688146341851072788944167206503055360.000000, -1942
24465011780785615866434981859602103818236871052068376623894611481072580188949932
073780677617280247166015796001340653568.000000]

Caso 3: Posición inicial [1000. 1000. 1000. 1000.]

[01-20 17:39:17] [INFO] i= 0 x: [1000. 1000. 1000. 1000.]

[01-20 17:39:17] [INFO] i= 1 x: [[360. -510. 1120. 3150.]]

[01-20 17:39:17] [INFO] i= 2 x: [[-1390. -8590. -1310. 14055.]]

[01-20 17:39:17] [INFO] i= 3 x: [[-4610. -46955. -22330. 61092.5]]

[01-20 17:39:17] [INFO] i= 4 x: [[-935. -205412.5 -141100. 239393.75]]

[01-20 17:39:17] [INFO] i= 5 x: [[78147.5 -780003.75 -695555.
824053.125]]

[01-20 17:39:17] [INFO] i= 6 x: [[612466.25 -2554118.125 -2953647.5
2356178.4375]]

[01-20 17:39:17] [INFO] i= 7 x: [[3354536.875 -6666515.9375 -11018061.25
4492568.28125]]

[01-20 17:39:17] [INFO] i= 8 x: [[15370966.5625 -9123669.53125 -35371684.375
-3998512.890625]]

[01-20 17:39:17] [INFO] i= 9 x: [[6.16210592e+07 3.82460435e+07 -8.89932378e+07
-1.01863859e+08]]

[01-20 17:39:17][INFO] i= 10 x: [[2.16233879e+08 4.32833349e+08
 -1.01496919e+08 -6.99996658e+08]]
 [01-20 17:39:17][INFO] i= 11 x: [[6.35828546e+08 2.63432273e+09
 6.62670331e+08 -3.62014711e+09]]
 [01-20 17:39:17][INFO] i= 12 x: [[1.30898343e+09 1.28320962e+10
 6.59398359e+09 -1.59511507e+10]]
 [01-20 17:39:17][INFO] i= 13 x: [[-3.55869618e+08 5.40915672e+10
 3.88521571e+10 -6.17112704e+10]]
 [01-20 17:39:17][INFO] i= 14 x: [[-2.36127456e+10 2.00373224e+11
 1.85887446e+11 -2.07616111e+11]]
 [01-20 17:39:17][INFO] i= 15 x: [[-1.71401667e+11 6.37334113e+11
 7.72521337e+11 -5.69740499e+11]]
 [01-20 17:39:17][INFO] i= 16 x: [[-9.07708559e+11 1.57403428e+12
 2.81971090e+12 -9.51195963e+11]]
 [01-20 17:39:17][INFO] i= 17 x: [[-4.06538752e+12 1.60791127e+12
 8.78749034e+12 1.98187827e+12]]
 [01-20 17:39:17][INFO] i= 18 x: [[-1.59670694e+13 -1.31252139e+13
 2.07908032e+13 3.00832224e+13]]
 [01-20 17:39:17][INFO] i= 19 x: [[-5.47068203e+13 -1.24165684e+14
 1.53311787e+13 1.93914116e+14]]
 [01-20 17:39:17][INFO] i= 20 x: [[-1.54828042e+14 -7.21239211e+14
 -2.17669011e+14 9.73024310e+14]]
 [01-20 17:39:17][INFO] i= 21 x: [[-2.85901188e+14 -3.42264313e+15
 -1.87781644e+15 4.19505647e+15]]
 [01-20 17:39:17][INFO] i= 22 x: [[3.32989757e+14 -1.41299961e+16
 -1.06009191e+16 1.58945346e+16]]
 [01-20 17:39:17][INFO] i= 23 x: [[7.07184219e+15 -5.12126807e+16
 -4.94618305e+16 5.20881058e+16]]
 [01-20 17:39:17][INFO] i= 24 x: [[4.77109803e+16 -1.58015168e+17
 -2.01349022e+17 1.36348240e+17]]
 [01-20 17:39:17][INFO] i= 25 x: [[2.44682877e+17 -3.65710866e+17
 -7.18728380e+17 1.89202109e+17]]
 [01-20 17:39:17][INFO] i= 26 x: [[1.07174589e+18 -2.14588813e+17
 -2.16887849e+18 -7.62556027e+17]]
 [01-20 17:39:17][INFO] i= 27 x: [[4.12316817e+18 4.24195776e+18
 -4.76693461e+18 -8.74640395e+18]]
 [01-20 17:39:17][INFO] i= 28 x: [[1.37758270e+19 3.52481042e+19
 -1.04995370e+18 -5.33971332e+19]]
 [01-20 17:39:17][INFO] i= 29 x: [[3.73480116e+19 1.96489457e+20
 6.83963010e+19 -2.60536036e+20]]
 [01-20 17:39:17][INFO] i= 30 x: [[5.96968554e+19 9.09701263e+20
 5.29771517e+20 -1.09966614e+21]]
 [01-20 17:39:17][INFO] i= 31 x: [[-1.49841771e+20 3.67892816e+21
 2.87894556e+21 -4.07891945e+21]]
 [01-20 17:39:17][INFO] i= 32 x: [[-2.07896297e+21 1.30367410e+22
 1.31157474e+22 -1.29972377e+22]]
 [01-20 17:39:17][INFO] i= 33 x: [[-1.31947539e+22 3.89127067e+22
 5.23049768e+22 -3.22165716e+22]]

[01-20 17:39:17][INFO] i= 34 x: [[-6.56972469e+22 8.32574448e+22
 1.82435367e+23 -3.36684837e+22]]
 [01-20 17:39:17][INFO] i= 35 x: [[-2.81613289e+23 1.82752901e+21
 5.31385623e+23 2.62951518e+23]]
 [01-20 17:39:17][INFO] i= 36 x: [[-1.06094372e+24 -1.31841265e+24
 1.06642630e+24 2.51083213e+24]]
 [01-20 17:39:17][INFO] i= 37 x: [[-3.45126526e+24 -9.91733533e+24
 -5.03972688e+23 1.46240167e+25]]
 [01-20 17:39:17][INFO] i= 38 x: [[-8.90938995e+24 -5.32854126e+25
 -2.08426160e+25 6.95068109e+25]]
 [01-20 17:39:17][INFO] i= 39 x: [[-1.16001805e+25 -2.40963229e+26
 -1.48256057e+26 2.87316815e+26]]
 [01-20 17:39:17][INFO] i= 40 x: [[5.55488854e+25 -9.54657617e+26
 -7.78438573e+26 1.04276714e+27]]
 [01-20 17:39:17][INFO] i= 41 x: [[6.02219529e+26 -3.30452046e+27
 -3.46619238e+27 3.22368450e+27]]
 [01-20 17:39:17][INFO] i= 42 x: [[3.62786430e+27 -9.50938160e+27
 -1.35414257e+28 7.49335955e+27]]
 [01-20 17:39:17][INFO] i= 43 x: [[1.75734698e+28 -1.84480346e+28
 -4.61016146e+28 4.62124456e+27]]
 [01-20 17:39:17][INFO] i= 44 x: [[7.37551946e+28 1.37898463e+28
 -1.29099298e+29 -8.52344186e+28]]
 [01-20 17:39:17][INFO] i= 45 x: [[2.71988443e+29 3.98592401e+29
 -2.30618904e+29 -7.13198053e+29]]
 [01-20 17:39:17][INFO] i= 46 x: [[8.59830208e+29 2.76880546e+30
 3.35946993e+29 -3.98523470e+30]]
 [01-20 17:39:17][INFO] i= 47 x: [[2.09691148e+30 1.43885626e+31
 6.20950491e+30 -1.84780914e+31]]
 [01-20 17:39:17][INFO] i= 48 x: [[1.96955274e+30 6.36133318e+31
 4.11961349e+31 -7.48219302e+31]]
 [01-20 17:39:17][INFO] i= 49 x: [[-1.87789381e+31 2.46882988e+32
 2.09618933e+32 -2.65515015e+32]]
 [01-20 17:39:17][INFO] i= 50 x: [[-1.72354879e+32 8.33809098e+32
 9.13003842e+32 -7.94211726e+32]]
 [01-20 17:39:17][INFO] i= 51 x: [[-9.92198586e+32 2.30344043e+33
 3.49362588e+33 -1.70834771e+33]]
 [01-20 17:39:17][INFO] i= 52 x: [[-4.68381133e+33 3.93485768e+33
 1.15941326e+34 -1.05220209e+32]]
 [01-20 17:39:17][INFO] i= 53 x: [[-1.92534076e+34 -7.34361432e+33
 3.10579806e+34 2.65444118e+34]]
 [01-20 17:39:17][INFO] i= 54 x: [[-6.94595755e+34 -1.18034830e+35
 4.74287326e+34 2.00766612e+35]]
 [01-20 17:39:17][INFO] i= 55 x: [[-2.12892295e+35 -7.67763398e+35
 -1.41212195e+35 1.08103900e+36]]
 [01-20 17:39:17][INFO] i= 56 x: [[-4.85339007e+35 -3.86966820e+36
 -1.81795119e+36 4.89552671e+36]]
 [01-20 17:39:17][INFO] i= 57 x: [[-2.33765827e+35 -1.67382971e+37
 -1.13752388e+37 1.94198263e+37]]

[01-20 17:39:17][INFO] i= 58 x: [[6.01218041e+36 -6.36225373e+37
 -5.62270718e+37 6.73202701e+37]]
 [01-20 17:39:17][INFO] i= 59 x: [[4.88316063e+37 -2.09356276e+38
 -2.39699218e+38 1.94184804e+38]]
 [01-20 17:39:17][INFO] i= 60 x: [[2.70042161e+38 -5.52211470e+38
 -8.98110988e+38 3.79261712e+38]]
 [01-20 17:39:17][INFO] i= 61 x: [[1.24401051e+39 -7.91885618e+38
 -2.90064492e+39 -2.62494031e+38]]
 [01-20 17:39:17][INFO] i= 62 x: [[5.00940422e+39 2.89624139e+39
 -7.38506107e+39 -8.03689262e+39]]
 [01-20 17:39:17][INFO] i= 63 x: [[1.76663635e+40 3.43919803e+40
 -8.97763935e+39 -5.60767902e+40]]
 [01-20 17:39:17][INFO] i= 64 x: [[5.23472590e+40 2.11599990e+41
 5.08286820e+40 -2.91985644e+41]]
 [01-20 17:39:17][INFO] i= 65 x: [[1.09942626e+41 1.03672824e+42
 5.24857344e+41 -1.29266369e+42]]
 [01-20 17:39:17][INFO] i= 66 x: [[-1.29864474e+40 4.38986197e+42
 3.12317117e+42 -5.02320736e+42]]
 [01-20 17:39:17][INFO] i= 67 x: [[-1.85648038e+42 1.63363129e+43
 1.50260663e+43 -1.69914362e+43]]
 [01-20 17:39:17][INFO] i= 68 x: [[-1.37158197e+43 5.22845552e+43
 6.27247583e+43 -4.70644536e+43]]
 [01-20 17:39:17][INFO] i= 69 x: [[-7.31649615e+43 1.30753158e+44
 2.30018627e+44 -8.11204231e+43]]
 [01-20 17:39:17][INFO] i= 70 x: [[-3.29284096e+44 1.44095800e+44
 7.21543569e+44 1.44628085e+44]]
 [01-20 17:39:17][INFO] i= 71 x: [[-1.29899134e+45 -1.01133202e+45
 1.73127874e+45 2.38263741e+45]]
 [01-20 17:39:17][INFO] i= 72 x: [[-4.47388950e+45 -9.89052298e+45
 1.43989343e+45 1.55557312e+46]]
 [01-20 17:39:17][INFO] i= 73 x: [[-1.27703098e+46 -5.79976100e+46
 -1.69012591e+46 7.85457854e+46]]
 [01-20 17:39:17][INFO] i= 74 x: [[-2.41950918e+46 -2.76733707e+47
 -1.49797738e+47 3.40201692e+47]]
 [01-20 17:39:17][INFO] i= 75 x: [[2.28617692e+46 -1.14754104e+48
 -8.53062890e+47 1.29478012e+48]]
 [01-20 17:39:17][INFO] i= 76 x: [[5.58584737e+47 -4.17881851e+48
 -4.00120787e+48 4.26762384e+48]]
 [01-20 17:39:17][INFO] i= 77 x: [[3.82359722e+48 -1.29804822e+49
 -1.63600528e+49 1.12906968e+49]]
 [01-20 17:39:17][INFO] i= 78 x: [[1.97396234e+49 -3.04925199e+49
 -5.86810698e+49 1.63982450e+49]]
 [01-20 17:39:17][INFO] i= 79 x: [[8.68696197e+49 -2.10061851e+49
 -1.78347180e+50 -5.76643122e+49]]
 [01-20 17:39:17][INFO] i= 80 x: [[3.35688174e+50 3.30333931e+50
 -3.98706729e+50 -6.94854261e+50]]
 [01-20 17:39:17][INFO] i= 81 x: [[1.12774739e+51 2.81360344e+51
 -1.36745597e+50 -4.28877796e+51]]

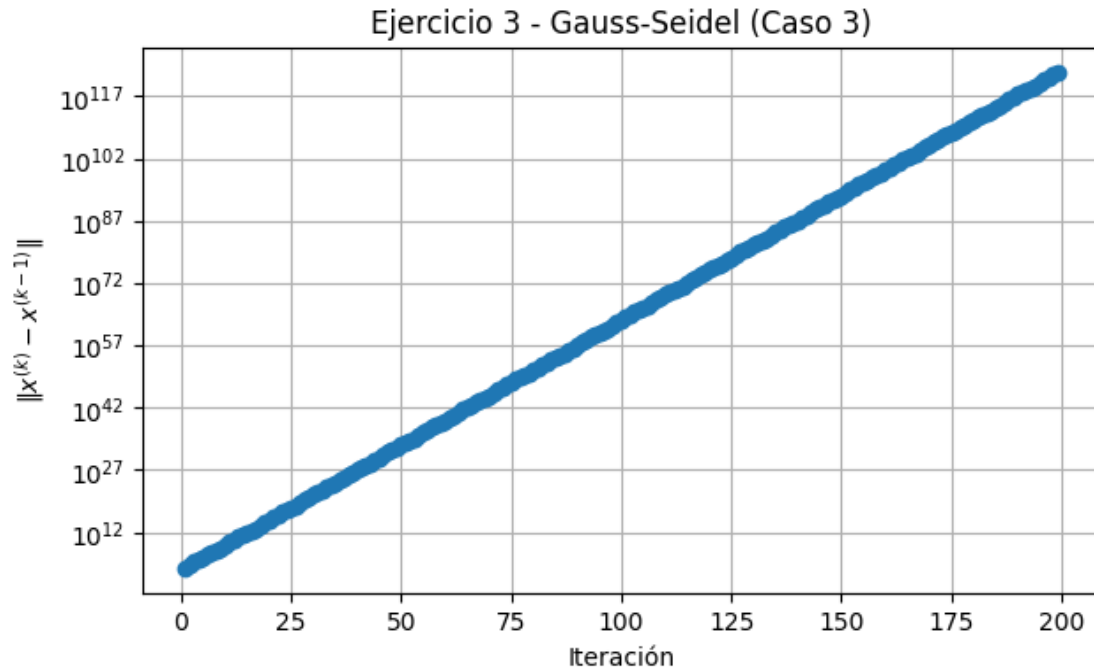
[01-20 17:39:17][INFO] i= 82 x: [[3.08709464e+51 1.58166829e+52
 5.35371569e+51 -2.10481665e+52]]
 [01-20 17:39:17][INFO] i= 83 x: [[5.10925154e+51 7.36074669e+52
 4.23407972e+52 -8.92408017e+52]]
 [01-20 17:39:17][INFO] i= 84 x: [[-1.10741276e+52 2.98989075e+53
 2.31896528e+53 -3.32535348e+53]]
 [01-20 17:39:17][INFO] i= 85 x: [[-1.64803982e+53 1.06469859e+54
 1.06177121e+54 -1.06616228e+54]]
 [01-20 17:39:17][INFO] i= 86 x: [[-1.05884382e+54 3.20141423e+54
 4.25293959e+54 -2.67565155e+54]]
 [01-20 17:39:17][INFO] i= 87 x: [[-5.30446495e+54 6.97542929e+54
 1.49087076e+55 -3.00879012e+54]]
 [01-20 17:39:17][INFO] i= 88 x: [[-2.28419860e+55 1.09309200e+54
 4.37682739e+55 2.02444989e+55]]
 [01-20 17:39:17][INFO] i= 89 x: [[-8.64434558e+55 -1.03408679e+56
 8.97227318e+55 1.99974384e+56]]
 [01-20 17:39:17][INFO] i= 90 x: [[-2.82854142e+56 -7.93054562e+56
 -2.73718939e+55 1.17589590e+57]]
 [01-20 17:39:17][INFO] i= 91 x: [[-7.38310775e+56 -4.29337036e+57
 -1.64085291e+57 5.61962908e+57]]
 [01-20 17:39:17][INFO] i= 92 x: [[-1.01166453e+57 -1.95114047e+58
 -1.18684465e+58 2.33328838e+58]]
 [01-20 17:39:17][INFO] i= 93 x: [[4.22548839e+57 -7.76416094e+58
 -6.27597025e+58 8.50825629e+58]]
 [01-20 17:39:17][INFO] i= 94 x: [[4.78777955e+58 -2.70129596e+59
 -2.80802624e+59 2.64793082e+59]]
 [01-20 17:39:17][INFO] i= 95 x: [[2.91475652e+59 -7.83706217e+59
 -1.10186444e+60 6.24627106e+59]]
 [01-20 17:39:17][INFO] i= 96 x: [[1.42002266e+60 -1.55572310e+60
 -3.77114131e+60 4.48013988e+59]]
 [01-20 17:39:17][INFO] i= 97 x: [[5.98655953e+60 8.71376251e+59
 -1.06537288e+61 -6.63392879e+60]]
 [01-20 17:39:17][INFO] i= 98 x: [[2.21788339e+61 3.14268914e+61
 -1.95647051e+61 -5.69226897e+61]]
 [01-20 17:39:17][INFO] i= 99 x: [[7.05563017e+61 2.21759666e+62
 2.37243726e+61 -3.20777312e+62]]
 [01-20 17:39:17][INFO] i= 100 x: [[1.74310920e+62 1.16036723e+63
 4.90968076e+62 -1.49506681e+63]]
 [01-20 17:39:17][INFO] i= 101 x: [[1.78431077e+62 5.15459957e+63
 3.30267061e+63 -6.08056405e+63]]
 [01-20 17:39:17][INFO] i= 102 x: [[-1.45074165e+63 2.00936211e+64
 1.69145404e+64 -2.16831615e+64]]
 [01-20 17:39:17][INFO] i= 103 x: [[-1.37354596e+64 6.82285652e+64
 7.40163230e+64 -6.53346863e+64]]
 [01-20 17:39:17][INFO] i= 104 x: [[-7.98040807e+64 1.90216301e+65
 2.84489776e+65 -1.43079564e+65]]
 [01-20 17:39:17][INFO] i= 105 x: [[-3.78763251e+65 3.34965216e+65
 9.49412155e+65 -2.77417463e+64]]

[01-20 17:39:17][INFO] i= 106 x: [[-1.56385909e+66 -5.31221700e+65
 2.56875474e+66 2.08120992e+66]]
 [01-20 17:39:17][INFO] i= 107 x: [[-5.66873118e+66 -9.34360621e+66
 4.07506608e+66 1.60529424e+67]]
 [01-20 17:39:17][INFO] i= 108 x: [[-1.74937384e+67 -6.15774993e+67
 -1.05370802e+67 8.70977089e+67]]
 [01-20 17:39:17][INFO] i= 109 x: [[-4.05033389e+67 -3.12333546e+68
 -1.44229159e+68 3.96385739e+68]]
 [01-20 17:39:17][INFO] i= 110 x: [[-2.38752274e+67 -1.35726160e+69
 -9.13125410e+68 1.57932970e+69]]
 [01-20 17:39:17][INFO] i= 111 x: [[4.68989216e+68 -5.18212530e+69
 -4.54077403e+69 5.50280093e+69]]
 [01-20 17:39:17][INFO] i= 112 x: [[3.89942276e+69 -1.71497541e+70
 -1.94457986e+70 1.60017318e+70]]
 [01-20 17:39:17][INFO] i= 113 x: [[2.17418432e+70 -4.57091507e+70
 -7.31911054e+70 3.19681734e+70]]
 [01-20 17:39:17][INFO] i= 114 x: [[1.00673060e+71 -6.84225654e+70
 -2.37800512e+71 -1.62664080e+70]]
 [01-20 17:39:17][INFO] i= 115 x: [[4.07178459e+71 2.18177171e+71
 -6.12446155e+71 -6.33488834e+71]]
 [01-20 17:39:17][INFO] i= 116 x: [[1.44306948e+72 2.73108983e+72
 -7.88537969e+71 -4.49090373e+72]]
 [01-20 17:39:17][INFO] i= 117 x: [[4.30816577e+72 1.69923390e+73
 3.88510372e+72 -2.35459566e+73]]
 [01-20 17:39:17][INFO] i= 118 x: [[9.22213154e+72 8.37451051e+73
 4.17548854e+73 -1.04740215e+74]]
 [01-20 17:39:17][INFO] i= 119 x: [[2.35334288e+71 3.56210865e+74
 2.50999981e+74 -4.08816306e+74]]
 [01-20 17:39:17][INFO] i= 120 x: [[-1.45789097e+74 1.33165980e+75
 1.21442169e+75 -1.39027886e+75]]
 [01-20 17:39:17][INFO] i= 121 x: [[-1.09718358e+75 4.28807469e+75
 5.09216299e+75 -3.88603053e+75]]
 [01-20 17:39:17][INFO] i= 122 x: [[-5.89625129e+75 1.08540033e+76
 1.87604753e+76 -6.90076728e+75]]
 [01-20 17:39:17][INFO] i= 123 x: [[-2.66669474e+76 1.27958298e+76
 5.92289573e+76 1.04207339e+76]]
 [01-20 17:39:17][INFO] i= 124 x: [[-1.05662085e+77 -7.76953293e+76
 1.44049574e+77 1.88567781e+77]]
 [01-20 17:39:17][INFO] i= 125 x: [[-3.65794478e+77 -7.87448247e+77
 1.32708490e+77 1.24752661e+78]]
 [01-20 17:39:17][INFO] i= 126 x: [[-1.05286523e+78 -4.66273658e+78
 -1.30947951e+78 6.33936512e+78]]
 [01-20 17:39:17][INFO] i= 127 x: [[-2.04377755e+78 -2.23713524e+79
 -1.19444322e+79 2.75848125e+79]]
 [01-20 17:39:17][INFO] i= 128 x: [[1.51751197e+78 -9.31813578e+79
 -6.86315692e+79 1.05456252e+80]]
 [01-20 17:39:17][INFO] i= 129 x: [[4.40817806e+79 -3.40918545e+80
 -3.23625854e+80 3.49564890e+80]]

[01-20 17:39:17][INFO] i= 130 x: [[3.06333163e+80 -1.06598736e+81
 -1.32908880e+81 9.34436644e+80]]
 [01-20 17:39:18][INFO] i= 131 x: [[1.59219023e+81 -2.54020849e+81
 -4.79015232e+81 1.41523658e+81]]
 [01-20 17:39:18][INFO] i= 132 x: [[7.04009614e+81 -1.99576592e+81
 -1.46607216e+82 -4.33671193e+81]]
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 -1.52757672e+82 -3.44381045e+83]]
 [01-20 17:39:18][INFO] i= 135 x: [[2.55046975e+83 1.27291434e+84
 4.18439347e+83 -1.70015184e+84]]
 [01-20 17:39:18][INFO] i= 136 x: [[4.36035648e+83 5.95493052e+84
 3.38270738e+84 -7.24104209e+84]]
 [01-20 17:39:18][INFO] i= 137 x: [[-8.10484243e+83 2.42953494e+85
 1.86752758e+85 -2.71053862e+85]]
 [01-20 17:39:18][INFO] i= 138 x: [[-1.30552022e+85 8.69362322e+85
 8.59412504e+85 -8.74337231e+85]]
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 3.45754965e+86 -2.22066744e+86]]
 [01-20 17:39:18][INFO] i= 140 x: [[-4.28213779e+86 5.83741418e+86
 1.21810223e+87 -2.66561011e+86]]
 [01-20 17:39:18][INFO] i= 141 x: [[-1.85246305e+87 1.65322219e+86
 3.60368730e+87 1.55386032e+87]]
 [01-20 17:39:18][INFO] i= 142 x: [[-7.04205239e+87 -8.09994605e+87
 7.53801904e+87 1.59189286e+88]]
 [01-20 17:39:18][INFO] i= 143 x: [[-2.31759841e+88 -6.33947509e+88
 -1.12385402e+87 9.45301993e+88]]
 [01-20 17:39:18][INFO] i= 144 x: [[-6.11470428e+88 -3.45861495e+89
 -1.29037210e+89 4.54273637e+89]]
 [01-20 17:39:18][INFO] i= 145 x: [[-8.77870752e+88 -1.57964520e+90
 -9.49797409e+89 1.89456909e+90]]
 [01-20 17:39:18][INFO] i= 146 x: [[3.19949622e+89 -6.31355506e+90
 -5.05888521e+90 6.94088998e+90]]
 [01-20 17:39:18][INFO] i= 147 x: [[3.80421537e+90 -2.20773398e+91
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 [01-20 17:39:18][INFO] i= 148 x: [[2.34124213e+91 -6.45631675e+91
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 [01-20 17:39:18][INFO] i= 149 x: [[1.14725714e+92 -1.30986320e+92
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 -8.78803072e+92 -5.15321460e+92]]
 [01-20 17:39:18][INFO] i= 151 x: [[1.80821943e+93 2.47538074e+93
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 1.63800231e+93 -2.58143133e+94]]
 [01-20 17:39:18][INFO] i= 153 x: [[1.44795384e+94 9.35604807e+94
 3.87870906e+94 -1.20947176e+95]]

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 2.31625427e+97 -1.19698739e+97]]
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 7.77266124e+97 -3.80845760e+96]]
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 2.12348910e+98 1.62954551e+98]]
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 [01-20 17:39:18][INFO] i= 163 x: [[-2.33503638e+099 -1.10040034e+101
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 [01-20 17:39:18][INFO] i= 164 x: [[3.65198769e+100 -4.22020296e+101
 -3.66639978e+101 4.49710456e+101]]
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 -1.57732055e+102 1.31810726e+102]]
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 2.01672589e+106 -3.32664331e+106]]
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 4.13333367e+107 -3.20718240e+107]]
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 1.52984663e+108 -5.85693636e+107]]
 [01-20 17:39:18][INFO] i= 176 x: [[-2.15928196e+108 1.12764558e+108
 4.86051587e+108 7.38789564e+107]]
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[01-20 17:39:18][INFO] i= 178 x: [[-2.99018848e+109 -6.26616216e+109
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 6.95512446e+117 -7.16837174e+117]]
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 2.96604374e+119 1.18859729e+119]]
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 -4.07704140e+122 5.66124553e+122]]



Solución encontrada: [24098668696430998777632954758510463059195193154767176279588457402104265948476429713167857165476283391562847000046042349568.000000, -513317748997822899452959492648390419438712468943440296008031895306443734196509227815675013455245836749916259053096319057920.000000, -407704140368215721650791673284061085410745667996547525380810056131743119878191559251776225563458124541950712312163704242176.000000, 566124553312626488354043402330555086452695869416886681321642814893794041355668062097624407401139692853899032423562626465792.000000]

Animaciones:

Las animaciones se encuentran en el repositorio en la carpeta animations..